Optimal Screening under Entry: Quantity Caps and Endogenous Market Segmentation

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Abstract

We study how entry reshapes monopoly screening under asymmetric information. In our model, an incumbent with full contractual flexibility moves first, followed by an entrant restricted to linear pricing. The firms compete to serve a downstream buyer with privately known willingness to pay and inelastic demand. Although the incumbent can offer an arbitrarily complex menu, its optimal mechanism consists of a single quantity-payment option. We establish this result in two steps. Starting from an arbitrary menu, we first construct a class of alternative menus that preserve the entrant's best response and use an extreme-point argument to show that the incumbent can restrict attention to menus with at most two options. We then reduce any such two-option menu to a single option by reallocating quantity from lower- to higher-valuation types. The incumbent's optimal quantity-payment option can be implemented through a quantity cap and a per-unit price below the entrant's. The cap allows the incumbent to target intermediate-valuation types without competing directly for high-valuation types, who instead purchase from the entrant, thereby endogenously segmenting the market. Relative to the monopoly benchmark, entry reduces the incumbent's profit while increasing both consumer surplus and total surplus.

1 Introduction

How does an incumbent monopolist respond to entry when the buyer is privately informed? Although monopoly pricing under asymmetric information is well understood (see, e.g.,

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Mussa and Rosen (1978); Maskin and Riley (1984); Wilson (1993)), the monopolist's response to entry remains less well understood. Unlike in the monopoly benchmark, the incumbent's offer now influences the entrant's subsequent pricing decision. The incumbent's contract therefore serves two roles: extracting surplus from the buyer and shaping the entrant's residual demand. This strategic interaction makes contract design substantially more challenging than in the standard monopoly screening problem.

We focus on a market in which an incumbent supplies an intermediate input to a downstream buyer with inelastic demand and faces entry by a rival firm. The incumbent has greater pricing flexibility: the entrant is restricted to linear pricing, whereas the incumbent can offer arbitrary quantity-payment pairs. Absent entry, the incumbent's optimal mechanism is simply a take-it-or-leave-it linear price. With entry, however, its broader contract space may allow it to soften direct price competition and deter aggressive undercutting by the entrant.

Such contractual flexibility is common in practice. Supply contracts often feature nonlinear pricing or minimum- and maximum-volume provisions. In manufacturing and government procurement, contracts may specify a maximum quantity that a supplier is obligated to provide over a given period, with additional requirements met through renegotiation or alternative suppliers. Similar arrangements arise in energy markets, where long-term contracts cover a base quantity and spot markets accommodate residual demand. These provisions can endogenously segment procurement between a contracted supplier and its rivals, although the competitive effects of this segmentation remain largely unexplored.

This issue is particularly relevant when established firms have access to a broader set of pricing instruments than entrants and other competitors, an asymmetry observed in several antitrust cases. Microsoft, for example, used nonlinear licensing contracts, while its main competitors relied on per-unit pricing (see, e.g., Baseman, Warren-Boulton, and Woroch (1995)). Similarly, in the Tomra and Tetra Pak cases, the incumbents employed sophisticated pricing schemes, including rebates and discounts contingent on purchase volumes, that their competitors did not match (see, e.g., Fu and Tan (2019); Chao, Tan, and Wong (2022)). These examples motivate our analysis of how an incumbent can use contractual flexibility to defend part of its market against a rival with a simpler pricing instrument.[1]

To answer this question, we propose a three-stage model with two firms and a privately informed buyer. The firms sequentially offer exclusive contracts for a homogeneous divisible good with a fixed total quantity. Firm 1 moves first and can choose a flexible nonlinear tariff

[1]This motive reflects a broader concern that disruptive entrants can change the incumbent's competitive environment and eventually erode its market position–a concern captured by Jensen Huang's remark: "I don't want to let someone come in and be the price leaders, lock me out at the bottom, and climb their way to the top" (Tae Kim, *The Nvidia Way*, p. 116).

which induces a menu of quantity-payment pairs, whereas Firm 2 observes Firm 1's offer and then chooses a linear price. After observing both offers, the buyer chooses from which firm to purchase, or takes the outside option. Our analysis proceeds in two steps. First, we solve the optimal contract design problem: despite Firm 1's ability to offer an arbitrary menu, we show that an optimal menu can be reduced to a single quantity-payment option. Second, we establish that this single option can be implemented as linear pricing with a quantity cap, and then study how this capped offer shapes the equilibrium in response to entry.

We begin by representing Firm 1's menu by the incentive-compatible allocation and transfer rule it induces. The buyer's utility is linear in quantity, and therefore Firm 2's linear pricing offer is equivalent to a posted price for the full unit. This observation also explains why Firm 1 cannot simply mimic Firm 2's full-unit offer. If Firm 1 allocated the full unit to a positive mass of types, Firm 2 would have an undercutting incentive to capture those types. Hence, without loss, Firm 1's allocation can be taken to be strictly below the full unit almost everywhere. This implies that the slope of the buyer's payoff from Firm 1 with respect to the buyer's type—which equals the allocation received by that type—is strictly smaller than the corresponding slope of the buyer's payoff from Firm 2. Consequently, the two payoff functions satisfy a single-crossing property in type. The buyer's behavior is therefore summarized by a single marginal type: lower trading types choose Firm 1, while higher types choose Firm 2. Equivalently, given Firm 1's allocation, Firm 2's pricing problem can be written as a choice of this marginal type.

The main difficulty is Firm 1's problem, since its strategy space is large and a change in its allocation generally changes Firm 2's best response. Rather than solving this infinite-dimensional problem directly, we use a dominance argument. Given any allocation, we look for another allocation that gives Firm 1 weakly higher profit while inducing Firm 2 to choose the same marginal type. The key is that Firm 2's incentive at the margin depends only on the marginal type's payoff from Firm 1 and Firm 1's allocation near that type. Fixing these objects defines a class of allocations within which Firm 1 can rearrange allocations among inframarginal types without affecting Firm 2's best response. Optimizing over this class yields weakly higher profit than the given allocation. Since only the allocation below the marginal type matters for Firm 1's profit, we truncate these allocations at the marginal type and obtain a linear problem. By Bauer's Maximum Principle (Bauer (1958)), the objective attains its maximum at an extreme point of the resulting feasible set of truncated allocations. Using the characterization in Kleiner, Moldovanu, and Strack (2021), we show that such extreme points have a simple structure. This first reduces Firm 1's problem to menus with at most two quantity-payment options (Proposition 1).

We then show that any such two-option menu can be optimally reduced to a single-option

menu. The argument is constructive. Starting from a two-option allocation, we reallocate quantity from lower to higher types while keeping the area under the allocation and its level near the margin fixed, so that Firm 2's best response remains the same. Under regularity, we show that the gain from shifting quantity toward higher-valuation types dominates the loss from shifting quantity away from lower-valuation types. The resulting allocation has only one nonzero quantity level. Thus, Firm 1 can restrict attention to a single quantity-payment option without loss (Proposition 2).

The intuition is that a rich screening menu is unnecessary once competition is summarized by a single marginal type. What matters is how Firm 1's offer shapes this margin through Firm 2's pricing incentive. Once this competitive margin is fixed, additional options below the margin mainly refine screening within Firm 1's segment and raise information rents, rather than strengthening Firm 1's competitive position. Hence, Firm 1 can optimally respond to entry by concentrating its offer on a single quantity-payment pair, rather than using a complex menu to screen all buyer types within its segment.

We next use the single-option result to reduce the equilibrium analysis to a finite-dimensional problem. Since the buyer's demand is inelastic, a single quantity-payment option can be implemented as linear pricing with a quantity cap: Firm 1 charges a unit price and limits the quantity the buyer can purchase from Firm 1. The cap is the key instrument through which Firm 1 differentiates its offer from Firm 2's full-unit option. A larger cap makes the two offers closer substitutes, which strengthens Firm 2's incentive to respond with a lower price and induces a lower marginal type. Firm 1 thus faces a tradeoff: a larger cap increases the quantity sold to each served buyer, but it also intensifies competition from Firm 2 and reduces the segment served by Firm 1.

We establish the existence of an equilibrium by showing that there exists an optimal single option that is implemented as a capped offer (Theorem 1). In equilibrium, Firm 1 offers this capped option at a lower unit price than Firm 2's posted price. The lower unit price is necessary because the cap is less than the full unit; otherwise, every type would prefer Firm 2's full-unit offer. At the same time, the cap is what makes the discount sustainable. By capping the quantity available under its offer, Firm 1 makes its discounted option attractive to intermediate types while limiting direct competition with Firm 2 for high types, who value the full unit more. The equilibrium is therefore best understood as competitive market segmentation: low types do not trade, intermediate types buy Firm 1's capped option, and high types buy Firm 2's full-unit option. Under a sufficient condition, we show that the equilibrium is essentially unique: all equilibria induce the same marginal type and the same single-option allocation for Firm 1 almost everywhere below that type (Proposition 3).

The equilibrium market segmentation also has important implications for profits and

welfare. Although Firm 1's contractual flexibility allows it to remain active and earn positive profit, Firm 2 earns more by selling the full unit to high types. Relative to the monopoly benchmark, entry pushes both equilibrium unit prices below the monopoly price and reduces industry profit. It also expands the set of served types. In particular, even Firm 2's upper segment contains all types served under monopoly, while Firm 1 serves lower types who would otherwise remain unserved. This market expansion raises total surplus and, together with the lower equilibrium prices, increases buyer surplus. Thus, competition does not merely lower prices relative to monopoly; it changes the nature of the incumbent's problem from screening a buyer within a single firm to segmenting demand across firms.

We illustrate the equilibrium characterization through two tractable examples. With uniformly distributed buyer valuations, we derive a closed-form equilibrium by sequentially optimizing Firm 1's unit price and quantity cap. We then consider a family of beta distributions with linear virtual valuations to examine how changes in the distribution of buyer valuations affect market outcomes. Although an increase in the beta parameter shifts the distribution toward lower valuations in the sense of first-order stochastic dominance, Firm 1 can initially benefit as greater heterogeneity allows it to attract more intermediate types and increase its profit. We also show that a higher common marginal cost raises both firms' unit prices and reduces their demands and profits, while leaving Firm 1's optimal cap unchanged.

Our analysis applies to markets for large language models and other compute-intensive AI services. Paid subscription plans often include usage limits, while API access is commonly priced per token or model call. These arrangements resemble Firm 1's capped offer and Firm 2's linear pricing in our model. When customers differ in their willingness to pay for additional usage, a discounted capped plan may attract intermediate-valuation customers, while high-valuation customers may prefer a rival's service without such limits. Our analysis suggests that usage caps can differentiate competing offers, soften price competition, and segment the market.

The U.S. airline industry also provides a useful illustration of incumbent responses to entry. During the 1990s and early 2000s, Southwest entered routes already served by incumbent airlines and used a relatively simple pricing strategy.[2] Goolsbee and Syverson (2008) show that incumbent fares fall as the threat of Southwest entry increases, with much of the decline occurring before Southwest begins service. Prince and Simon (2015) find that incumbents' on-time performance also deteriorates in response to Southwest entry and entry threats, and discuss product differentiation as one possible explanation, given Southwest's superior on-time performance. Taken together, these findings resemble the response in our model:

[2]Southwest described its pricing as a "relatively simple fare structure." See Southwest Airlines Co., *2004 Form 10-K*, U.S. Securities and Exchange Commission, available at `https://www.sec.gov/Archives/edgar/data/92380/000095013405002093/d21965e10vk.htm`.

the incumbent lowers its price while differentiating its offer rather than simply matching the entrant's offer.

Related Literature. Our analysis connects to the classic literature on monopoly screening and nonlinear pricing. Mussa and Rosen (1978) and Maskin and Riley (1984) provide canonical analyses of optimal monopoly tariffs under asymmetric information, where buyers self-select among menu options and incentive constraints generate information rents and downward distortions for lower types. Stokey (1979) and Anderson and Dana (2009) show that discrimination need not be profitable when the buyer's utility is linear in the allocation; the monopolist's optimal pricing scheme may remain simple rather than take the form of a rich menu. Our paper contributes to this line of work by showing that a similar simplicity result can survive under competition with linear buyer utility. In our model, the incumbent can offer an arbitrary menu, yet the equilibrium outcome can still be implemented by a simple contract rather than a rich screening schedule.[3]

This article provides a theoretical benchmark for competition among firms facing a buyer with incomplete information. Recent papers study simultaneous menu competition explicitly. Chade and Swinkels (2021) study screening in vertical oligopolies with menu competition and characterize the resulting equilibrium sorting and distortions. Kojima and Pakzad-Hurson (2025) show that even when firms compete through menus, Bertrand logic can remain powerful: competition drives profits to zero and yields efficient allocations. More broadly, related work shows that the contractual environment can shape competition under asymmetric information. Participation choices, exclusivity provisions, and contingent terms can affect competitive pressure and participation distortions (Martimort and Stole (2009); Calzolari and Denicolò (2013)). Calzolari and Denicolò (2015) show that, when a dominant firm has a competitive advantage, exclusive contracts can leverage buyers' information rents to partially foreclose rivals. While these papers focus on simultaneous competition, we study a sequential setting and examine how an established firm uses its richer contract space in response to entry.

Our paper is closely related to Chao, Tan, and Wong (2018) and Chao, Tan, and Wong (2022), which also consider similar settings with two asymmetric firms moving sequentially. However, they assume complete information on the demand side, which is different from our model. Both papers derive the optimal pricing scheme for the dominant firm when facing a

[3]A less direct connection is to Halac, Lipnowski, and Rappoport (2026), where pricing instruments are also used to shape a strategic environment. The connection to our paper is that, in both settings, pricing affects not only demand directly but also the strategic environment in which demand is determined. The difference is that their mechanism operates through discount dispersion and equilibrium selection under network externalities, whereas ours operates through competitive screening against a rival with a simpler pricing instrument.

follower with a restricted pricing strategy. In particular, using a mechanism design approach, Chao, Tan, and Wong (2022) show that the optimal nonlinear pricing strategy of the first mover is convex while exhibiting quantity discounts. Moreover, it involves uncountably many unchosen pairs that constrain the follower's profitable pricing deviations.

The strategic use of contracts in our setting also connects to the literature on contractual entry deterrence and exclusion. Aghion and Bolton (1987) show that liquidated damages in an incumbent-buyer contract can act as an entry fee that the entrant must effectively cover to attract the buyer. As a result, the contract can block some efficient entrants without fully precluding entry. Related work on exclusive dealing studies how incumbent contracts can limit an entrant's access to buyers and affect entry (Rasmusen, Ramseyer, and Wiley (1991); Bernheim and Whinston (1998); Segal and Whinston (2000); Fumagalli and Motta (2006)). Ide, Montero, and Figueroa (2016) study whether discounts and rebates can play a similar exclusionary role. The common theme is that the incumbent's contract changes the rival's post-entry opportunity. The difference lies in how this effect operates. In our model, the incumbent's contract does not serve to exclude the entrant; instead, it uses a quantity cap to shape the entrant's residual demand and segment the market.

One of our main findings is the presence of a second-mover advantage. Existing literature provides conditions on the reaction curve that determine whether a first-mover or second-mover advantage will emerge. Gal-Or (1985) focuses on symmetric firms, while Dowrick (1986) extends the analysis to asymmetric firms. Adopting a different notion of advantage, Amir and Stepanova (2006) derive related conditions based on the underlying demand functions.[4] Additionally, Amir and Grilo (1999) endogenize the order of play by allowing players to decide whether to move early or late before engaging in either a Cournot or Stackelberg game. They also identify the conditions on the inverse demand function under which simultaneous or sequential play arises in subgame perfect equilibria.

A recent literature in mechanism design studies how a designer's contract or mechanism interacts with an aftermarket or outside option that is not under the designer's control (Dworczak (2020)). Like our paper, this literature shares the feature that the designer (the first mover) must anticipate the impact of its strategy on the aftermarket. For example, Hendricks and Wiseman (2022) study a sequential setting in which the first seller designs a mechanism while anticipating a fixed second-stage selling procedure, so the second-stage option constrains the first seller's contract. Loertscher and Muir (2022) analyze a monopolist's optimal randomization strategy (e.g., through rationing or opaque pricing) in the presence of resale. They show that resale can reduce the monopolist's gains from random-

[4]They define two distinct games based on which firm moves first and posit that a firm has a first-mover (or second-mover) advantage if its equilibrium payoff is higher when it moves first (or second) compared to the alternative game.

ization and, counterintuitively, harm consumer surplus when it distorts the seller's optimal quantity decision. Kang and Muir (2022) examine a vertically integrated platform that sells inputs upstream and competes downstream with fringe producers. They endogenize downstream competition and solve for the optimal nonlinear menu offered by the platform. Finally, Kang (2023) considers a public option in a private market, where the policymaker's choice of allocation and quality for the public option affects equilibrium prices in the private sector. They show that optimal rationing and distortion of public quality arise due to pecuniary externalities. The key difference is that our model allows the follower to be fully strategic, whereas these approaches typically impose parametric assumptions or rely on market-clearing conditions in the aftermarket.

Methodologically, our analysis is related to a literature that uses extreme-point arguments to characterize the structure and complexity of solutions to infinite-dimensional design problems. Manelli and Vincent (2007) formulate revenue maximization as a linear problem over the convex set of incentive-compatible mechanisms and study the extreme points of this set. Kleiner, Moldovanu, and Strack (2021) characterize extreme points of monotone functions under majorization constraints, and Nikzad (2023) studies such problems with additional constraints and derives bounds on the complexity of optimal mechanisms. Yang and Zentefis (2024) characterize extreme points of monotone function intervals. More recently, Lahr and Niemeyer (2025) show that, with multidimensional types, extreme points of incentive-compatible mechanisms are pervasive and generically have uncountable menus. Yang and Yang (2025) characterize extreme points of multidimensional monotone functions, with applications to mechanism design and information design. In our application, holding the entrant's best response fixed yields a majorization set generated by a step function, so an optimal extreme-point allocation has at most two options.

The remainder of the paper is organized as follows. In Section 2, we introduce the model. Section 3 studies the optimal contract design problem and establishes the optimality of a single-option menu. Section 4 characterizes the quantity-cap equilibrium and derives its properties. Section 5 discusses several extensions and directions for future research. Section 6 concludes the article. All omitted proofs are collected in the appendices.

2 Model

There are two upstream firms that produce a homogeneous intermediate good and compete to supply a downstream buyer. The buyer's payoff from consuming $q \in [0, Q_0]$ units of the good and making a payment t is $\theta q - t$, where θ is her privately known willingness to

pay.[5] The buyer's type θ is distributed according to a C^3 cumulative distribution function F with full support on $\Theta = [0,1]$ and a strictly positive density f. Throughout the paper, we assume that F is regular: its virtual valuation, $J(\theta) := \theta - \frac{1-F(\theta)}{f(\theta)}$, satisfies $J'(\theta) > 0$ for every $\theta \in (0,1)$.

For each $i \in \{1,2\}$, Firm i produces quantity q at cost $c_i q$, where $c_i \in [0,1)$ is its constant marginal cost. We initially assume symmetric costs, $c_1 = c_2 = c$. Each firm offers an exclusive contract, so the buyer may purchase from at most one firm.[6] Firm 1 may offer a flexible nonlinear tariff, whereas Firm 2 is restricted to a linear tariff, that is, a constant per-unit price. Because the buyer's utility is linear in quantity, any type that purchases from Firm 2 optimally buys the full unit. Firm 2's linear tariff is therefore equivalent to a posted price for one unit, which we denote by $p_2 \in \mathbb{R}_+$. We describe Firm 1's contract space in greater detail below.

We consider the following sequential move game. At the outset, Nature draws the buyer's type θ from F, and only the buyer observes its realization. The game then proceeds as follows: (i) Firm 1 offers a contract; (ii) after observing Firm 1's contract, Firm 2 posts a price; and (iii) after observing both offers, the buyer purchases from the firm whose offer gives her higher payoff, or takes the outside option and receives zero payoff. If the buyer is indifferent between the two firms and obtains a strictly positive payoff, the tie is broken in favor of Firm 2. If neither offer yields a positive payoff, the buyer takes the outside option.

We focus on pure-strategy weak perfect Bayesian equilibrium (weak PBE) in which Firm 2's belief about θ remains equal to F after every offer by Firm 1, including off the equilibrium path. Under this belief restriction, the game has the same strategy and payoff implications as an equivalent formulation in which θ is drawn after Firm 2 posts its price. The restriction reflects the fact that Firm 1 does not observe θ when choosing its contract, so its offer conveys no information about the buyer's type.[7] Hence, Firm 2 faces the same pricing problem under either timing, and the buyer's purchase decision and Firm 1's optimization problem are unchanged. We therefore work with the equivalent formulation and characterize its subgame perfect equilibrium (SPE), which also characterizes the weak PBE of the original formulation under the stated belief restriction.

We now formally define the contract space used by Firm 1. Firm 1 chooses a nonlinear

[5] Because the buyer's payoff is linear in q, we normalize $Q_0 = 1$ without loss of generality. Our main results also extend to the case in which the buyer's utility is $\theta v(q) - t$, where v is convex. See Section 5 for details.

[6] This assumption captures single sourcing in settings where using multiple suppliers is costly. For example, enterprise customers may rely on a single software or IT provider because combining multiple systems can involve integration and compatibility costs. Similarly, manufacturers may rely on one supplier for specialized components when inputs from different suppliers are not easily compatible.

[7] This is the no-signaling-what-you-don't-know condition of Fudenberg and Tirole (1991).

tariff $T : [0,1] \to \mathbb{R}$, where $T(q)$ specifies the total payment for quantity q. Let

$$\mathcal{T} := \{\, T : [0,1] \to \mathbb{R} : T \text{ is lower semicontinuous} \,\}$$

denote the set of tariffs from which Firm 1 chooses. This class includes, for example, a linear tariff $T(q) = pq$; a two-part tariff with $T(0) = 0$, $T(q) = A + pq$ for $q > 0$, and $A \geq 0$; and a linear tariff with a quantity cap $\hat{q} \in (0,1)$, which sets $T(q) = pq$ for $q \leq \hat{q}$ and $T(q) = M$ for $q > \hat{q}$, where M is sufficiently large that no buyer type chooses a quantity above $\hat{q}$.

Each tariff induces a menu of quantity-payment pairs.[8] Given $T \in \mathcal{T}$, define the associated menu[9]

$$C_T := \{(q, T(q)) : \ q \in [0,1]\} \cup \{(0,0)\}.$$

For each $\theta \in \Theta$, the buyer of type θ chooses an option in C_T that maximizes her utility. Since $[0,1]$ is compact and T is lower semicontinuous, a maximizer exists, and we can choose

$$\big(q_T(\theta), t_T(\theta)\big) \in \arg\max_{(q,t)\in C_T} \{\theta q - t\},$$

which specifies the quantity and payment selected by type θ under the menu C_T. Define the associated indirect utility by $u_T(\theta) := \max_{(q,t)\in C_T}\{\theta q - t\} = \theta q_T(\theta) - t_T(\theta)$ for $\theta \in \Theta$.[10]

We can thus equivalently view Firm 1 as choosing a menu C_T induced by some tariff $T \in \mathcal{T}$ from $\mathcal{C} := \{C_T : T \in \mathcal{T}\}$. Facing menu C_T, type θ buyer chooses the option $\big(q_T(\theta), t_T(\theta)\big)$ and obtains $u_T(\theta)$.[11]

Remark 1. Given that Firm 1 offers a flexible contract, one could also allow its menu to depend on Firm 2's subsequently posted price. We exclude such price-contingent contracts because they would give Firm 1 an additional commitment instrument: by specifying different terms for different prices, Firm 1 could directly affect Firm 2's pricing incentives. Our interest is instead in how Firm 1 uses contractual flexibility while committing to a menu before Firm 2 moves. We therefore focus on contracts posted independently, as is standard

[8] Throughout, we use the terms contract and menu interchangeably when referring to Firm 1's offer.

[9] It is without loss to include the null option $(0,0)$ in C_T, since the buyer can always take the outside option.

[10] In case of a tie where the buyer is indifferent between distinct options in Firm 1's menu, the tie-breaking rule does not matter. Such a tie can occur only at a type where u_T is not differentiable. Since u_T is convex, there are at most countably many such types, so the choice among tied options does not affect either firm's payoff.

[11] The restriction to lower semicontinuous tariffs is without loss of generality among tariffs for which the buyer's problem admits a maximizer. For any such tariff T, let T^{lsc} denote its lower semicontinuous envelope (i.e., the largest lower semicontinuous function below T). Since $T^{\text{lsc}} \leq T$, $u_{T^{\text{lsc}}}(\theta) \geq u_T(\theta)$. Conversely, for any q, there exists a sequence $q_n \to q$ such that $T(q_n) \to T^{\text{lsc}}(q)$, so $\theta q - T^{\text{lsc}}(q) \leq u_T(\theta)$. Hence $u_{T^{\text{lsc}}} = u_T$. Since this common indirect utility is convex, the chosen quantity and payment coincide almost everywhere. Thus, replacing T by T^{lsc} does not affect either firm's payoff.

in models of competitive nonlinear pricing (Stole (1995); Armstrong and Vickers (2001); Rochet and Stole (2002)), with Firm 2 subsequently choosing its price against Firm 1's given menu.

A pure-strategy SPE is characterized by a profile consisting of Firm 1's strategy $C_T^* \in \mathcal{C}$, Firm 2's strategy $p^* : \mathcal{C} \to \mathbb{R}_+$, and the buyer's strategy $s^* = (s_0^*, s_1^*, s_2^*)$, where $s_i^* : \Theta \times \mathcal{C} \times \mathbb{R}_+ \to \{0,1\}$ indicates whether type θ purchases from Firm i, for $i \in \{1,2\}$, and $s_0^*(\theta; C_T, p_2)$ indicates whether she takes the outside option. If the buyer purchases from Firm 1, she chooses the option $(q_T(\theta), t_T(\theta))$ defined above. These strategies satisfy[12]

$$\begin{aligned} s_1^*(\theta; C_T, p_2) &= \mathbb{1}\{u_T(\theta) > \max\{0, \theta - p_2\}\}, \qquad s_2^*(\theta; C_T, p_2) = \mathbb{1}\{\theta - p_2 \geq u_T(\theta)\}, \\ s_0^*(\theta; C_T, p_2) &= 1 - s_1^*(\theta; C_T, p_2) - s_2^*(\theta; C_T, p_2), \qquad \forall (C_T, p_2) \in \mathcal{C} \times \mathbb{R}_+, \end{aligned} \tag{1}$$

$$p^*(C_T) \in \arg\max_{p_2 \in \mathbb{R}_+} (p_2 - c) \int_\Theta s_2^*(\theta; C_T, p_2)\, dF(\theta), \quad \forall C_T \in \mathcal{C}, \tag{2}$$

$$C_T^* \in \arg\max_{C_T \in \mathcal{C}} \int_\Theta s_1^*(\theta; C_T, p^*(C_T)) \left[t_T(\theta) - c\, q_T(\theta)\right] dF(\theta). \tag{3}$$

Given both firms' contracts, the buyer chooses the firm whose contract yields a higher payoff, or takes the outside option (Equation (1)). Firm 2 chooses an optimal price conditional on Firm 1's contract while taking into account the buyer's behavior (Equation (2)).[13] Finally, Firm 1 chooses an optimal contract anticipating Firm 2's and the buyer's behavior (Equation (3)). Hereafter, equilibrium denotes SPE. Our main interest is in characterizing Firm 1's optimal contract and providing comparative statics analysis of the equilibrium.

3 Optimal Contract Design

3.1 Equivalent Representations

Our analysis proceeds by rewriting Firm 1's contract choice in terms of the allocation and transfer it induces across types. This reformulation lets us frame Firm 1's strategy space as in a standard screening problem. Define

$$\mathcal{M} := \left\{\big(x(\cdot), t(\cdot)\big) : \Theta \to [0,1] \times \mathbb{R} \text{ satisfying (IC) and (IR)}\right\}$$

[12] For notational convenience, when type $\theta = p_2$ receives zero payoff from both firms, we assign this type to Firm 2 rather than to the outside option. This case involves only a single type and is therefore payoff-irrelevant.

[13] For any fixed C_T, Firm 2's demand $\int_\Theta \mathbf{1}\{\theta - u_T(\theta) \geq p_2\}\, dF(\theta)$ is well defined because $u_T(\theta)$ is continuous, and it is upper semicontinuous in p_2. Moreover, Firm 2's problem can be restricted to $[c, 1]$, on which its profit is upper semicontinuous and therefore attains a maximum.

where

$$\theta x(\theta) - t(\theta) \geq \theta x(\theta') - t(\theta'), \quad \text{for any } \theta, \theta' \in \Theta, \tag{IC}$$

$$\theta x(\theta) - t(\theta) \geq 0, \quad \text{for any } \theta \in \Theta. \tag{IR}$$

We show that the two representations are equivalent.

Lemma 1. *For any $C_T \in \mathcal{C}$, there exists $(x,t) \in \mathcal{M}$ such that $(q_T(\theta), t_T(\theta)) = (x(\theta), t(\theta))$ for every $\theta \in \Theta$. Conversely, for any $(x,t) \in \mathcal{M}$, there exists $C_T \in \mathcal{C}$ such that the same equality holds almost everywhere. Moreover, $u_T(\theta) = \theta x(\theta) - t(\theta)$ for every $\theta \in \Theta$.*

The last statement of the lemma shows that indirect utilities coincide under the two representations. Hence, for any price posted by Firm 2, the buyer makes the same choice between Firm 1, Firm 2, and the outside option. Firm 2 therefore faces the same pricing problem. In particular, any payoff for Firm 1 that can be expressed as the expectation of a measurable function of the type, allocation, transfer, and the follower's best response is the same under C_T and under (x,t). We therefore take Firm 1's strategy to be an allocation-transfer pair $(x,t) \in \mathcal{M}$.

As in the standard mechanism design literature, an allocation x is implementable if and only if $x \in \mathcal{X}$, where

$$\mathcal{X} := \{x(\cdot) : \Theta \to [0,1], \text{ non-decreasing}\}.$$

For any such allocation, the envelope theorem implies that the transfer is given, up to the constant $t(0)$, by

$$t(\theta) = t(0) + \theta x(\theta) - \int_0^\theta x(s)\, ds.$$

Hence, Firm 1's strategy reduces to choosing an allocation $x \in \mathcal{X}$, together with the constant $t(0)$. (IR) requires $t(0) \leq 0$. Moreover, we show that $t(0) = 0$ is optimal for Firm 1.[14]

3.2 Buyer's Problem

We first characterize the buyer's choice between the two firms' offers. We show that this choice is summarized by a unique marginal type z: types above z purchase from Firm 2, while types below z either purchase from Firm 1 or choose the outside option. This characterization

[14] See Appendix B for the proof. The same conclusion holds in the standard monopoly screening problem. Here, however, the argument is less immediate because changing $t(0)$ also affects Firm 2's pricing response. Intuitively, a subsidy raises the buyer's payoff from Firm 1 uniformly, including for types who receive zero quantity. Firm 1 can instead use its quantity terms to make its offer more attractive without incurring this uniform subsidy.

reduces the interaction between the firms to a single competitive margin and will be useful in analyzing both firms' problems.

Given Firm 1's offer (x,t) and Firm 2's price p_2, a buyer of type θ obtains payoffs $\int_0^\theta x(s)ds$ and $\theta - p_2$ from Firm 1 and Firm 2, respectively, whenever non-negative. We begin by showing that the equilibrium analysis can be restricted to a smaller class of allocations.

Lemma 2. *Fix an allocation $x \in \mathcal{X}$ offered by Firm 1 and a best response of Firm 2. There exists an allocation $x' \in \mathcal{X}$ such that $x'(\theta) < 1$ almost everywhere and Firm 1's profit under any best response by Firm 2 to x' is weakly higher than its profit under x and the original best response.*

If Firm 1 offers the full quantity to a positive measure of types, its offer is a perfect substitute for Firm 2's full-unit offer for those types. Firm 2 can then attract them by matching the payoff provided by Firm 1 and may lower its price further to expand demand. Lemma 2 shows that this full-quantity portion of Firm 1's allocation is unnecessary. Without loss of generality, Firm 1 can instead offer an allocation strictly below one almost everywhere, thereby differentiating its offer from Firm 2's full-unit offer and weakening Firm 2's incentive to undercut.

Accordingly, throughout the remainder of this section, we restrict attention to allocations x satisfying $x(\theta) < 1$ for almost every θ. Consequently, the difference between the payoff from Firm 2's full-unit offer and that from Firm 1's offer, $\int_0^\theta [1 - x(s)]ds - p_2$, is strictly increasing in θ, since its derivative is $1 - x(\theta) > 0$ almost everywhere. Hence, if $p_2 \le \int_0^1 [1 - x(s)]ds$, there is a unique $z \in [0,1]$ satisfying

$$p_2 = \int_0^z [1 - x(s)]\, ds. \tag{4}$$

Types above z strictly prefer Firm 2, types below z choose Firm 1 whenever $\int_0^\theta x(s)\, ds > 0$, and otherwise choose the outside option; type z is indifferent between the two firms.[15] If instead $p_2 > \int_0^1 [1 - x(s)]ds$, we define $z = 1$ since in such case Firm 2 faces zero demand. We refer to z as the marginal type. When the offers are clear from context, we sometimes write z without explicitly indicating its dependence on x and p_2.

3.3 Firm 2's Problem

Given Firm 1's offer, Firm 2's problem is to choose p_2 to maximize $(p_2 - c)(1 - F(z))$ subject to (4). The preceding argument implies that there is a one-to-one correspondence between p_2 and $z \in [0,1]$ if $p_2 \le \int_0^1 [1 - x(s)]ds$. Within this range, a higher price corresponds to a

[15] If the marginal type receives zero payoff, then $z = p_2$.

higher marginal type and lower demand $1-F(z)$. Prices above this range yield zero demand and hence zero profit, and are represented by $z = 1$. Therefore, Firm 2 can equivalently choose the marginal type $z \in \Theta$, with profit

$$\pi_2(z|x) := \Big(\int_0^z [1 - x(s)]ds - c \Big)\big(1 - F(z)\big).$$

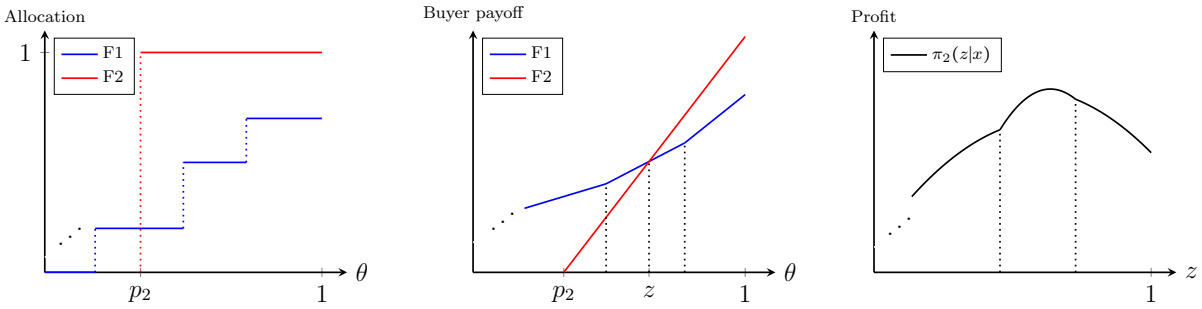


Figure 1: Given x (which is described as a step function in the left panel), each p_2 induces a unique marginal type z (the middle panel). The right panel describes Firm 2's profit as a function of z.

Observe that π_2 is continuous in z over Θ and differentiable almost everywhere. Whenever it is differentiable, its derivative with respect to z can be written as $\pi_2'(z|x) = -f(z)g(z|x)$, where

$$g(z|x) := \int_0^z [1 - x(s)]ds - c - \frac{1 - F(z)}{f(z)}\big(1 - x(z)\big).$$

Thus, the sign of π_2' is entirely determined by g.[16]

Lemma 3. *$g(\cdot|x)$ is strictly increasing on Θ for any $x \in \mathcal{X}$. Hence, $\pi_2(\cdot|x)$ is single-peaked with a unique maximizer, and any stationary point is necessarily the maximizer.*

We denote Firm 2's unique best response to x by z_x.[17] We next identify a class of alternative allocations that leave this marginal type unchanged. This class will also be useful in solving Firm 1's problem, as it separates how Firm 1's menu shapes competition at the margin from how it screens types within its own segment. To formalize this idea, given $x \in \mathcal{X}$ and the induced z_x, define

$$\Delta_x := \Big\{ \tilde{x}(\cdot) \in \mathcal{X} \;\Big|\; \int_0^{z_x} \tilde{x}(s)ds = \int_0^{z_x} x(s)ds \quad \text{and} \quad \tilde{x}(\theta) = x(z_x) \;\text{ for }\; \theta \in (z_x, 1] \Big\}.$$

[16] At points where x has a jump, π_2 may fail to be differentiable, as illustrated in Figure 1. In that case, the one-sided derivatives exist because $\int_0^z x(s)ds$ is convex, and they are obtained by replacing $x(z)$ with the one-sided limits $x(z+)$ and $x(z-)$, which exist because x is non-decreasing.

[17] If $g(\cdot|x)$ is continuous at an interior z_x, then $g(z_x|x) = 0$. Otherwise, optimality requires $g(z_x - |x) \le 0 \le g(z_x + |x)$, but $g(z_x|x)$ need not equal zero.

Lemma 4. *Fix $x \in \mathcal{X}$ and let z_x denote Firm 2's best response to x. If x is continuous at z_x, then every $\tilde{x} \in \Delta_x$ induces the same best response of Firm 2.*

The intuition behind Lemma 4 is as follows. Given an allocation x, what matters for Firm 2's pricing decision is not the entire allocation schedule, but the competitive margin it faces. The first condition defining the set of equivalent allocations Δ_x ensures that the marginal type receives the same payoff from Firm 1 under every allocation in the set. Consequently, the price at which Firm 2 makes this type indifferent between the two firms is preserved. At this price, the slope of Firm 2's profit around z_x is determined by Firm 1's allocation near the margin. To see this, suppose for simplicity that $\tilde{x} \in \Delta_x$ is continuous at z_x and that z_x is interior. Then,

$$g(z_x|\tilde{x}) - g(z_x|x) = \frac{1 - F(z_x)}{f(z_x)}\big(\tilde{x}(z_x) - x(z_x)\big).$$

Since z_x is the best response to x and x is continuous at z_x, we have $g(z_x|x) = 0$. The second condition limits any increase in Firm 1's allocation near the margin relative to the original allocation: $\tilde{x}(z_x) = x(z_x)$. Hence, $g(z_x|\tilde{x}) = 0$, so Firm 2's local pricing incentives are preserved and z_x remains its unique best response. Firm 1 can therefore rearrange allocations among inframarginal types below z_x without altering Firm 2's best response.

This result is further illustrated in Figure 2. Given an allocation x and Firm 2's best response z_x, consider a single-step allocation $\tilde{x}$. Choose $\lambda \in (0, z_x)$ so that the areas under x and $\tilde{x}$ on $[0, z_x]$ are the same:

$$\int_0^{z_x} \tilde{x}(s)ds = (z_x - \lambda) \cdot x(z_x) = \int_0^{z_x} x(s)ds.$$

Then $\tilde{x}$ induces the same best response for Firm 2, $z_{\tilde{x}} = z_x$.

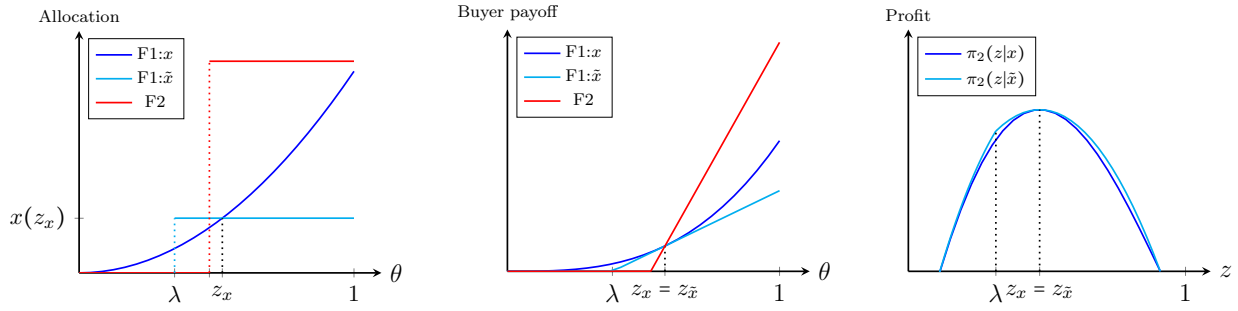


Figure 2: Illustration of Lemma 4.

3.4 Optimality of a Single-Option Menu

In this section, we characterize Firm 1's optimal contracts. Notice that Firm 1 chooses an allocation x to solve the following profit-maximization problem:

$$\max_{x\in\mathcal{X}} \int_0^{z_x} \left[(\theta - c)x(\theta) - \int_0^{\theta} x(s)ds\right] dF(\theta). \tag{5}$$

Integrating by parts, we can rewrite (5) as

$$\max_{x\in\mathcal{X}} \ \pi_1(x) = \int_0^{z_x} \left[\bar{J}(\theta|z_x) - c\right]x(\theta)\, dF(\theta), \tag{6}$$

where the *truncated virtual valuation function* is defined as

$$\bar{J}(\theta|z_x) := \theta - \frac{F(z_x) - F(\theta)}{f(\theta)}.$$

Unlike the monopoly problem, Firm 1's allocation affects its competitor's behavior through z_x, the unique maximizer of $\pi_2(\cdot|x)$. A change in the allocation affects Firm 1's expected profit both directly, through the profit it obtains from types below z_x, and indirectly, through Firm 2's response z_x. This indirect effect makes it challenging to solve (6).

In what follows, we adopt an approach involving a dominance argument and several transformations: given an arbitrary allocation x, we look for an allocation that induces the same best response to x from Firm 2 and produces a profit level that is weakly higher than that under x.

We proceed in three steps. In the first step, we utilize Lemma 4 and restrict Firm 1's maximization problem to Δ_x, the set of allocations that preserve Firm 2's best response to x.[18] Firm 1's restricted problem is

$$\max_{\tilde{x}\in\Delta_x} \int_0^{z_x} \left[\bar{J}(\theta|z_x) - c\right]\tilde{x}(\theta)\, dF(\theta). \tag{7}$$

Since Δ_x contains the allocation $x(\theta)\cdot\mathbb{1}\{\theta\in[0,z_x]\} + x(z_x)\cdot\mathbb{1}\{\theta\in(z_x,1]\}$, which coincides with x on the relevant region $[0,z_x]$, the maximum profit from (7) is weakly higher than $\pi_1(x)$. Therefore, solving (7) is sufficient for finding an allocation that yields weakly higher profit than x.[19]

[18] Without loss of generality, we restrict attention to baseline allocations x that are continuous at the induced best response z_x. See Lemma A.1.

[19] A natural approach to solving (7) is to formulate a Lagrangian. This approach, however, is complicated by the fact that the objective in (7) is integrated with respect to $dF(\theta)$, whereas the first constraint defining Δ_x is integrated with respect to $d\theta$. Consequently, the pointwise coefficient on the allocation takes the form $\left[\bar{J}(\theta|z_x) - c\right]f(\theta) - \mu$, where μ is the multiplier associated with the constraint. Pointwise maximization would

In our second step, we reformulate Firm 1's optimization problem (7) over Δ_x as a reduced problem over a truncated set Δ_x°, defined below, which keeps only the part of the allocation below the marginal type, the region relevant to Firm 1's profit. Specifically, define the set of truncated allocations associated with Δ_x by $\Delta_x^\circ := \{\tilde{x}|_{[0,z_x]} : \tilde{x} \in \Delta_x\}$ and consider the corresponding problem over Δ_x°:

$$\max_{\tilde{x}^\circ \in \Delta_x^\circ} \int_0^{z_x} \left[\bar{J}(\theta|z_x) - c\right] \tilde{x}^\circ(\theta)\, dF(\theta). \tag{8}$$

Every $\tilde{x} \in \Delta_x$ has a unique truncation $\tilde{x}^\circ \in \Delta_x^\circ$, and conversely every $\tilde{x}^\circ \in \Delta_x^\circ$ has a unique extension in Δ_x.[20] Under $\tilde{x}$ and its truncation $\tilde{x}^\circ$, the same types purchase from Firm 1. Therefore, (7) and (8) are equivalent in the sense that they deliver the same maximum profit and the same optimal allocation over the relevant types $[0, z_x]$.

The objective in (8) is linear and hence convex, while the choice set Δ_x° is convex and compact. By Bauer's Maximum Principle (Bauer (1958)), a convex upper-semicontinuous function on a nonempty compact convex subset of a Hausdorff locally convex space attains its maximum at an extreme point.

Next, we characterize the extreme points of Δ_x°. To do so, we introduce the standard definition of a mean-preserving spread and a well-known characterization of the extreme points of the mean-preserving spread set of a non-decreasing function.

Definition 1. *Let $\phi_1, \phi_2 : [0, \bar{\theta}] \to \mathbb{R}$ be non-decreasing. We say that ϕ_2 is a mean-preserving spread of ϕ_1, or $\phi_2 \in MPS(\phi_1)$, if*

$$\int_0^{\bar{\theta}} \phi_2(s)\, ds = \int_0^{\bar{\theta}} \phi_1(s)\, ds \quad \text{and} \quad \int_0^{\theta} \phi_2(s)\, ds \geq \int_0^{\theta} \phi_1(s)\, ds \quad \text{for all } \theta \in [0, \bar{\theta}].$$

Lemma 5 (Theorem 1 of Kleiner, Moldovanu, and Strack (2021))**.** *Let $\phi : [0, \bar{\theta}] \to \mathbb{R}$ be non-decreasing. Then $\hat{\phi}$ is an extreme point of $MPS(\phi)$ if and only if there exists a collection of disjoint intervals $[\underline{\theta}_i, \bar{\theta}_i)$ indexed by $i \in I$ such that, for a.e. $\theta \in [0, \bar{\theta}]$,*

$$\hat{\phi}(\theta) = \begin{cases} \phi(\theta), & \text{if } \theta \notin \bigcup_{i \in I} [\underline{\theta}_i, \bar{\theta}_i), \\ \dfrac{\int_{\underline{\theta}_i}^{\bar{\theta}_i} \phi(s)\, ds}{\bar{\theta}_i - \underline{\theta}_i}, & \text{if } \theta \in [\underline{\theta}_i, \bar{\theta}_i). \end{cases}$$

produce a single-option allocation if this coefficient had a unique zero and changed sign from negative to positive. Regularity ensures that $\bar{J}(\theta|z_x)$ is increasing, but it does not ensure the required sign pattern once the virtual surplus is weighted by $f(\theta)$. Establishing this pattern through a Lagrangian argument would therefore require a stronger restriction on the distribution. We take a different approach.

[20] For any $\tilde{x}^\circ \in \Delta_x^\circ$, define its extension $\tilde{x} \in \Delta_x$ on $[0,1]$ by setting $\tilde{x}(\theta) = \tilde{x}^\circ(\theta)$ for $\theta \in [0, z_x]$ and $\tilde{x}(\theta) = x(z_x)$ for $\theta \in (z_x, 1]$.

We show that the choice set Δ_x° can be represented as the mean-preserving spread set of a step function. Notice that we can equivalently write Δ_x° as

$$\Delta_x^\circ = \{\tilde{x}^\circ : [0, z_x] \to [0, x(z_x)] \,\big|\, \int_0^{z_x} \tilde{x}^\circ(s)ds = \int_0^{z_x} x(s)ds \quad \text{and} \quad \text{non-decreasing}\}.$$

This motivates us to define a step function $w : [0, z_x] \to [0, x(z_x)]$ by

$$w(\theta) = x(z_x) \cdot \mathbb{1}\{\theta \in [\lambda, z_x]\},$$

where λ is uniquely determined by $(z_x - \lambda)x(z_x) = \int_0^{z_x} x(s)\, ds$. We have the following result.

Lemma 6. $\Delta_x^\circ = MPS\,(w)$.

Interpreted as a cumulative distribution function after normalization, w represents a degenerate distribution concentrated at λ. Every feasible allocation corresponds to a distribution with the same mean and is therefore a mean-preserving spread of w. Given this representation of the choice set, by Lemma 5 with $\phi = w$ and $\bar{\theta} = z_x$, the extreme points of Δ_x° have a particularly simple structure. For any optimal extreme point $\hat{x}^\circ \in \Delta_x^\circ$, its extension $\hat{x} \in \Delta_x$ is an optimal solution to (7). The possible allocations are illustrated in Figure 3. Proposition 1 below takes the advantage of the simple structure of the extreme points.

Given an allocation-transfer pair (x, t), we call a pair (q, τ) with $q > 0$ an option if the set of types satisfying $(x(\theta), t(\theta)) = (q, \tau)$ has positive measure. We say that an allocation (or menu) has k options if there are exactly k such distinct pairs and almost every type is assigned one of them or the null option. For example, consider the allocation $\hat{x}$ in the top panel of Figure 3. When $\theta_1 < \lambda$, its positive allocation below z_x is strictly smaller than $x(z_x)$, so $\hat{x}$ is a two-option allocation. As $\theta_1 \uparrow \lambda$, this level converges to $x(z_x)$, yielding a single-option allocation at the limit.[21]

Proposition 1. *For any $x \in \mathcal{X}$, there exists $\hat{x}$ consisting of at most two quantity-payment options such that Firm 1's profit under $\hat{x}$ is weakly higher than under x.*

In our third step, we construct a transformation that reduces any relevant two-option allocation to a single option while strictly increasing Firm 1's profit. Proposition 2 then establishes that a single-option menu is optimal.

Proposition 2. *For any $x \in \mathcal{X}$, there exists a single-option allocation under which Firm 1's profit is weakly higher than under x. Furthermore, every optimal allocation has a single option almost everywhere below its induced marginal type.*

[21] This is exactly the extension of w.

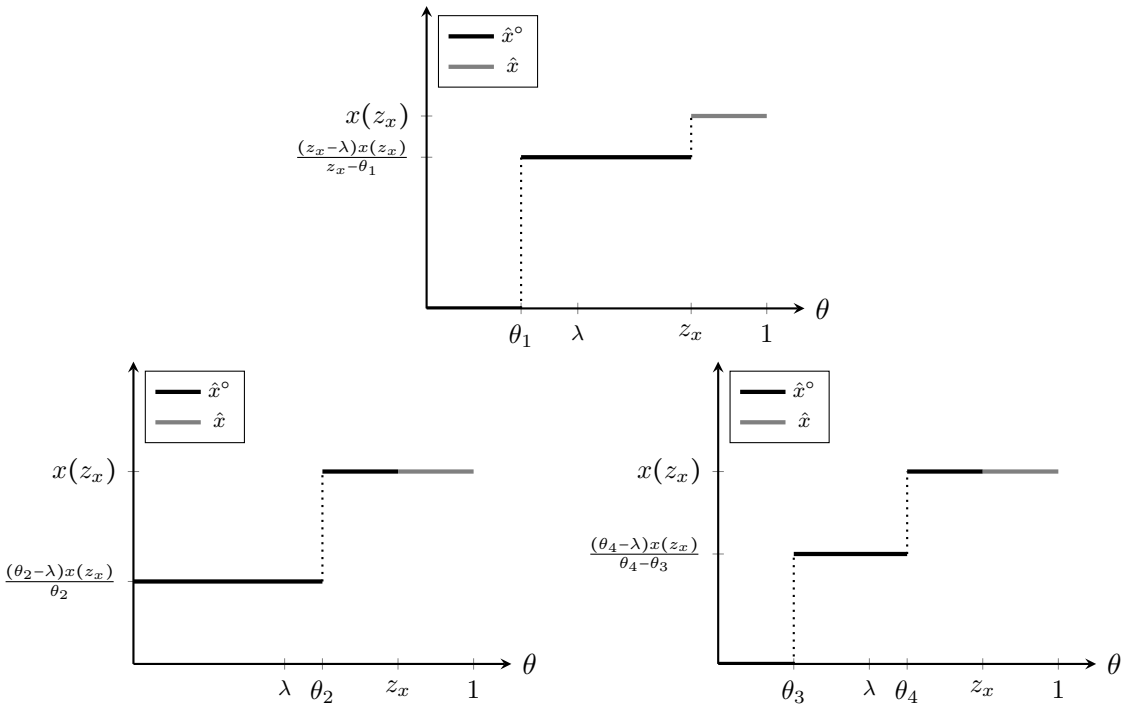


Figure 3: Extreme point $\hat{x}^\circ$ of Δ_x° and its extension $\hat{x} \in \Delta_x$.

To see how the reduction works, consider a candidate allocation $\hat{x} \in \Delta_x$ obtained in the proof of Proposition 1 and illustrated in Figure 3. We only need to consider the case in which $\hat{x}$ has two options. Let $\mathcal{X}^{(2)} \subset \mathcal{X}$ denote the class of two-option allocations. Any such candidate $\hat{x} \in \mathcal{X}^{(2)}$ can be represented by the pairs $\{(a_1, b_1), (a_2, b_2), (a_3, b_3)\}$, where $0 \leq a_1 < a_2 < a_3 = 1$, $0 = b_1 < b_2 < b_3$, and $b_i = \hat{x}(a_i)$ for $i \in \{1, 2, 3\}$. The case $a_1 = 0$ corresponds to the allocation in the bottom-left panel of Figure 3. Throughout, we focus on left-continuous allocations without loss of generality.[22]

Consider the following transformation that converts $\hat{x} \in \mathcal{X}^{(2)}$ into a single-option allocation $x^{(1)}$. Pick a point $\tilde{a} \in (a_1, a_2)$ and move types in $(a_1, \tilde{a}]$ from b_2 to 0 and types in $(\tilde{a}, a_2]$ from b_2 to b_3, preserving the area under the allocation: $\int_\Theta x^{(1)}(s)ds = \int_\Theta \hat{x}(s)ds$. This implies that such $\tilde{a}$ is unique and given by

$$\tilde{a} = \frac{a_1 b_2 + a_2(b_3 - b_2)}{b_3} \in (a_1, a_2).$$

Since $\hat{x} \in \Delta_x$, Firm 2's best response to $\hat{x}$ is z_x, and $a_2 \leq z_x$. The transformation changes the allocation only on $(a_1, a_2]$ while preserving its integral over this interval. Hence $x^{(1)} \in \Delta_x$, and by Lemma 4, Firm 2's best response remains z_x. The transformation is illustrated in Figure 4.

[22] We have the same results if we assume right-continuity.

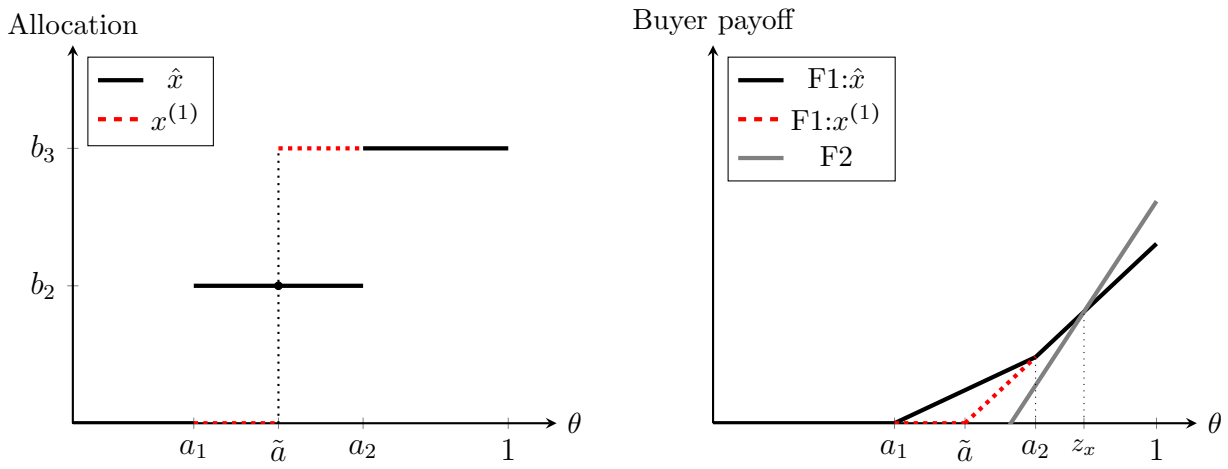


Figure 4: Transformation of $\hat{x}$ into $x^{(1)}$: allocation (left) and buyer's payoff (right). The transformation leaves Firm 2's best response unchanged.

Under regularity of the type distribution, we show that this transformation strictly raises Firm 1's profit: the gain from shifting quantity toward higher types dominates the loss from shifting quantity away from lower types. Hence any two-option candidate $\hat{x} \in \Delta_x$ can be optimally reduced to a single-option allocation.

The reason a rich menu is unnecessary is that competition is governed by a single margin. Given Firm 1's menu, Firm 2 chooses a posted price, or equivalently a marginal type who is indifferent between the two firms. Firm 1's menu therefore matters primarily through its effect on the competitive margin induced by Firm 2's best response. Lemma 4 shows that Firm 1 can rearrange its allocation while preserving this response. Once the marginal type is fixed, additional options only screen inframarginal types within Firm 1's segment. Under regularity, such screening is not profitable because it raises information rents without improving Firm 1's competitive position. Hence, despite its contractual flexibility, Firm 1 can optimally respond to entry with a single option rather than a rich screening menu.

Proposition 2 shows that Firm 1 can restrict attention to single-option menus, but it does not by itself guarantee the existence of an optimal menu. In the next section, we use this result to characterize the equilibrium.

4 The Equilibrium Quantity Cap, Endogenous Differentiation, and Market Segmentation

We now complete the characterization of the subgame-perfect equilibrium of the three-stage game by establishing its existence and uniqueness, examining its properties, and illustrating

the results through tractable examples.

4.1 Equilibrium Characterization

We represent Firm 1's single option by an allocation $x(\theta) = \hat{q}_1 \mathbb{1}\{\theta \geq p_1\}$, where p_1 is the lowest type willing to purchase the option and $\hat{q}_1 > 0$ is the quantity offered. Since this type is indifferent between purchasing and opting out, the total payment for the option is $p_1\hat{q}_1$. Because the buyer's utility is linear in quantity, this option can be implemented by charging a unit price p_1 and imposing a quantity cap $\hat{q}_1$. Any type purchasing from Firm 1 then buys the maximum available quantity $\hat{q}_1$. In equilibrium, the cap must satisfy $\hat{q}_1 < 1$, since otherwise Firm 2 can undercut Firm 1's full-unit offer. After observing Firm 1's capped offer $(p_1, \hat{q}_1)$, Firm 2 posts a price p_2 for the full unit.

We proceed by backward induction. Given $(p_1, \hat{q}_1)$ and p_2, the buyer obtains $(\theta - p_1)\hat{q}_1$ from buying Firm 1's capped option and $\theta - p_2$ from buying Firm 2's full-unit offer. The marginal type z satisfies $(z - p_1)\hat{q}_1 = z - p_2$, which is equivalent to

$$p_2 = (1 - \hat{q}_1)z + \hat{q}_1 p_1. \tag{9}$$

We focus on the case in which the marginal type is interior, $z \in (p_1, 1)$.[23] Since the difference between the two buying payoffs is strictly monotone in θ, the buyer's choice is characterized as follows: types in $[p_1, z)$ purchase from Firm 1, types in $[z, 1]$ purchase from Firm 2, and types below p_1 do not trade.

Using the one-to-one relation between p_2 and z in (9), Firm 2's profit can be written as

$$\pi_2(z|p_1, \hat{q}_1) = \big((1 - \hat{q}_1)z + \hat{q}_1 p_1 - c\big)\big(1 - F(z)\big).$$

The first order condition with respect to z gives

$$(1 - \hat{q}_1)J(z) + \hat{q}_1 p_1 = c. \tag{10}$$

Since the virtual valuation function J is strictly increasing, (10) uniquely defines the marginal type induced by Firm 2's best response to $(p_1, \hat{q}_1)$. We denote this type by $z(p_1, \hat{q}_1)$.

A cap changes the nature of competition. When $\hat{q}_1 < 1$, Firm 1's offer is not a perfect substitute for Firm 2's full-unit offer. The cap therefore creates quantity-based differentiation between the two offers: intermediate types may prefer Firm 1's capped option, while high types may still prefer paying more to obtain the full unit from Firm 2.

[23] We verify below that this indeed holds in equilibrium.

This differentiation affects Firm 2's pricing incentive. For a given p_2, a larger cap raises the payoff from Firm 1's offer for every type above p_1. The marginal type therefore moves upward, reducing Firm 2's demand. Combining (9) and (10) gives

$$p_2 - c = (1 - \hat{q}_1)\frac{1 - F(z)}{f(z)}.$$

Thus, Firm 2's markup is determined by the inverse hazard rate at the marginal type, scaled by $1 - \hat{q}_1$. A larger cap makes the two offers closer substitutes, increasing the competitive pressure on Firm 2. Indeed, implicit differentiation of (10) with respect to $\hat{q}_1$ gives

$$\frac{\partial z(p_1, \hat{q}_1)}{\partial \hat{q}_1} = \frac{c - p_1}{(1 - \hat{q}_1)^2 J'(z)} < 0.$$

Along Firm 2's best response, a larger cap therefore induces Firm 2 to charge a lower price and serve a lower marginal type.

We now return to Firm 1's problem. Since Firm 1 sells quantity $\hat{q}_1$ to types in $[p_1, z)$, its profit is

$$\pi_1(p_1, \hat{q}_1) = \hat{q}_1(p_1 - c)\big[F(z(p_1, \hat{q}_1)) - F(p_1)\big].$$

We consider how Firm 1's per-unit price and quantity cap affect its profit. A higher p_1 increases Firm 1's per-unit margin but reduces the mass of types it serves, both by raising the lowest type served and by lowering the marginal type through (10).

For fixed p_1, a larger cap makes Firm 1's offer closer to Firm 2's full-unit offer. As shown above, this induces Firm 2 to set a lower price and reduces the marginal type. Hence, a larger cap increases the quantity sold per buyer but shrinks Firm 1's intermediate segment. This makes Firm 1's profit generally non-monotone in the cap.

Firm 1 therefore jointly chooses p_1 and $\hat{q}_1$, balancing the pricing tradeoff and the differentiation effect described above. Anticipating Firm 2's response, Firm 1 solves

$$\max_{0<\hat{q}_1<1,\ c<p_1<z<1} \hat{q}_1(p_1 - c)\big[F(z) - F(p_1)\big] \quad \text{subject to} \quad (10). \tag{11}$$

We show that there exists a solution to (11), which yields the following equilibrium characterization.

Theorem 1. *There exists a pure-strategy subgame perfect equilibrium whose equilibrium path is characterized as follows:*

(i) Firm 1 offers the allocation rule

$$x^*(\theta) = \hat{q}_1^* \mathbb{1}\{\theta \geq p_1^*\}, \quad \theta \in [0, 1],$$

with the associated transfer rule

$$t^*(\theta) = p_1^* \hat{q}_1^* \mathbb{1}\{\theta \geq p_1^*\}.$$

(ii) Firm 2 charges

$$p_2^* = (1 - \hat{q}_1^*) z^* + \hat{q}_1^* p_1^*.$$

(iii) A buyer of type θ purchases from Firm 1 if $\theta \in (p_1^*, z^*)$, purchases from Firm 2 if $\theta \in [z^*, 1]$, and chooses the outside option if $\theta \in [0, p_1^*]$.

The equilibrium values p_1^*, $\hat{q}_1^*$, and z^* are determined by (11).

Proposition 2 shows that, if an optimal menu exists, Firm 1 can restrict attention to a single quantity-payment option. Theorem 1 completes the equilibrium construction by establishing existence of an equilibrium of the three-stage game and showing that the single option can be implemented as linear pricing together with a quantity cap. The cap allows Firm 1 to target intermediate types without competing directly for high types, who value the full unit more. The resulting equilibrium segments the market by type: low types choose the outside option, intermediate types purchase Firm 1's capped option, and high types purchase the full unit from Firm 2.

We next provide a sufficient condition under which (11) has a unique solution.

Proposition 3. *Suppose that $F(J^{-1}(\cdot))$ is weakly concave on $(J(c), c)$. Then (11) has a unique solution $(p_1^*, \hat{q}_1^*, z^*)$. All equilibria induce the marginal type z^* and the same single-option allocation for Firm 1 almost everywhere below z^*.*

Proposition 3 establishes that the equilibrium characterized in Theorem 1 is essentially unique: the equilibrium values of p_1^*, $\hat{q}_1^*$, p_2^*, and z^*, and hence the realized allocation and market segmentation, are uniquely determined. However, this does not imply uniqueness of Firm 1's entire menu. Starting from the single-option equilibrium, Firm 1 can modify its allocation above the marginal type without changing Firm 2's best response. Since these types purchase from Firm 2, such modifications generate alternative equilibrium menus without affecting the realized allocation or either firm's profit.

To prove uniqueness of the solution to (11), we solve the problem sequentially. For each fixed cap, we first optimize over p_1, with the marginal type z determined by (10). A higher p_1 lowers z, shrinking Firm 1's segment at both ends. The condition in Proposition 3 ensures that the marginal loss of types to Firm 2 does not diminish as p_1 increases. Consequently, Firm 1's profit is strictly concave in p_1, yielding a unique optimal price for each cap.

We then substitute this optimal price into Firm 1's profit and optimize over the cap. As the cap increases, both the optimal price and the induced marginal type decrease under stronger competition. A larger cap allows Firm 1 to sell more quantity to each type, but the lower price reduces its margin, while the condition implies that the competitive effect on demand does not weaken as the marginal type falls. Consequently, Firm 1's marginal profit decreases with the cap, and its profit evaluated at the optimal price is strictly concave. This establishes the uniqueness of the solution to (11). We illustrate this two-step maximization procedure with the uniform distribution in Section 4.3.

4.2 Properties of the Equilibrium

We now examine the properties of the equilibrium. In particular, we characterize how entry by the follower affects market outcomes, including prices, profits, buyer surplus, and total surplus. As a benchmark, consider the same setting without the follower. The monopolist optimally posts a single price p^m for the good, or equivalently uses linear pricing without a quantity cap, where p^m maximizes $\pi^m(p;c) := (p-c)\big(1-F(p)\big)$. We denote the corresponding monopoly profit by $\Pi^m = \pi^m(p^m;c)$, and denote buyer surplus and total surplus under the monopoly outcome by BS^m and TS^m, respectively. For the duopoly outcome, we omit superscripts on welfare objects.

Proposition 4. *The following results hold.*

(i) $p_1^* < p_2^* < z^* < p^m$.

(ii) $0 < \Pi_1^* < \Pi_2^*$.

(iii) $\Pi_1^* + \Pi_2^* < \Pi^m$, $BS^m < BS$, $TS^m < TS$.

Firm 1's capped option comes with a lower unit price, $p_1^* < p_2^*$; without this discount, Firm 2's full-unit offer would dominate Firm 1's offer for every type. The cap makes this discount sustainable by differentiating Firm 1's offer from Firm 2's full-unit offer. This allows Firm 1 to retain a positive measure of intermediate types and make positive profit.

The proposition also highlights a second-mover advantage. The presence of the follower forces Firm 1 to commit to a capped offer in order to remain active in the market, while Firm 2 serves the upper segment with its full-unit offer. This explains why the follower earns more even though the first mover has the more flexible contract space.

Finally, entry pushes both equilibrium unit prices below the monopoly price and expands the overall served segment from $[p^m, 1]$ under monopoly to $[p_1^*, 1]$ under duopoly. Moreover, even Firm 2's segment, $[z^*, 1]$, contains all types served under monopoly. Thus, the expansion in trade strictly increases total surplus and, together with the lower equilibrium prices,

also strictly increases buyer surplus. Industry profit, however, decreases because competition lowers both firms' margins.

4.3 Example

This subsection presents two tractable examples. The uniform distribution case yields a closed-form solution and illustrates how sequential optimization establishes equilibrium uniqueness. We then study comparative statics within a family of distributions with linear virtual valuations, focusing on how changes in the distribution of buyer valuations affect Firm 1's optimal offer and market outcomes.

Suppose that the buyer's type is uniformly distributed and the marginal cost is $c = 0$.[24] Then Firm 2's best-response condition (10) gives

$$z(p_1, \hat{q}_1) = \frac{1 - \hat{q}_1(1 + p_1)}{2(1 - \hat{q}_1)}.$$

Substituting this expression into Firm 1's objective, the profit-maximization problem (11) becomes

$$\max_{p_1, \hat{q}_1} \quad \frac{\hat{q}_1}{2}\left(p_1 - \frac{2 - \hat{q}_1}{1 - \hat{q}_1}p_1^2\right) \quad \text{s.t.} \quad 0 < \hat{q}_1 < 1, \;\; 0 < p_1 < \frac{1 - \hat{q}_1}{2 - \hat{q}_1}.$$

For any $\hat{q}_1 \in (0, 1)$, the objective is strictly concave in p_1, with a unique maximizer satisfying the constraint. The optimal price and corresponding marginal type are

$$p_1^*(\hat{q}_1) = \frac{1 - \hat{q}_1}{2(2 - \hat{q}_1)}, \quad z(p_1^*(\hat{q}_1), \hat{q}_1) = \frac{4 - 3\hat{q}_1}{4(2 - \hat{q}_1)}.$$

Both are strictly decreasing in the cap. To understand this, note that for any fixed cap,

$$z(p_1, \hat{q}_1) - p_1 = \frac{1}{2} - \frac{2 - \hat{q}_1}{2(1 - \hat{q}_1)}p_1. \tag{12}$$

A larger cap makes Firm 1's offer less differentiated from Firm 2's full-unit offer. As a result, the mass of types served by Firm 1 becomes more sensitive to its unit price, inducing Firm 1 to lower its optimal price. The linearity of (12) in p_1 also implies that the mass of types served at the optimal price is independent of the cap:

$$z(p_1^*(\hat{q}_1), \hat{q}_1) - p_1^*(\hat{q}_1) = \frac{1}{4}.$$

Thus, the marginal type falls by the same amount as the optimal price. A larger cap shifts

[24] Both regularity and the condition in Proposition 3 are satisfied, since $J'(\theta) = 2 > 0$ and $F(J^{-1}(y)) = (y + 1)/2$ is linear.

Firm 1's segment toward lower types while increasing the quantity sold to each type.

Firm 1's problem then reduces to

$$\max_{0<\hat{q}_1<1} \quad \frac{\hat{q}_1(1-\hat{q}_1)}{8(2-\hat{q}_1)}.$$

The objective is strictly concave in $\hat{q}_1$ and is uniquely maximized at $\hat{q}_1^* = 2-\sqrt{2}$, yielding the equilibrium outcomes

$$(p_1^*, \hat{q}_1^*, z^*, p_2^*) = \left(\frac{2-\sqrt{2}}{4}, 2-\sqrt{2}, \frac{3-\sqrt{2}}{4}, \frac{1}{4}\right).$$

Figure 5 shows the corresponding equilibrium allocation and market segmentation.

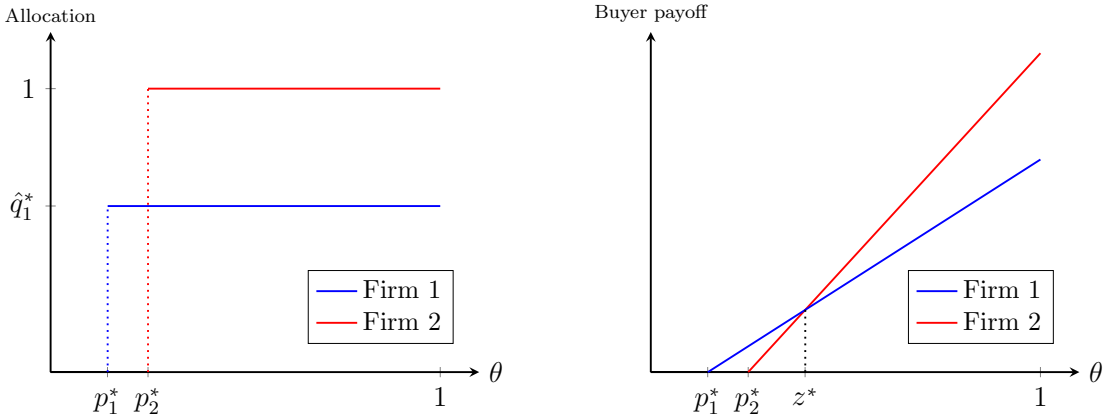


Figure 5: Equilibrium allocation (left) and buyer's payoff (right) when $\theta \sim U[0,1]$ and $c = 0$.

We now extend the analysis to the family of distributions with a linear virtual valuation $J(\theta)$, which remains tractable for comparative statics. This class corresponds to $\theta \sim \text{Beta}(\alpha, \beta)$ with $\alpha = 1$ and $\beta > 0$, whose CDF is $F_\beta(\theta) = 1-(1-\theta)^\beta$.[25] The case $\beta = 1$ gives the uniform distribution.

The density $f_\beta(\theta)$ is increasing in θ for $0 < \beta < 1$ and decreasing for $\beta > 1$, placing more weight on higher and lower valuations, respectively. As β increases, the distribution shifts toward lower valuations in the sense of first-order stochastic dominance: $F_{\beta'}(\theta) > F_\beta(\theta)$ for all $\theta \in (0,1)$ whenever $\beta' > \beta$. Consequently, the mean decreases with β, while the variance first increases and then decreases. As $\beta \to 0$ and $\beta \to \infty$, the distribution converges to point masses at 1 and 0, respectively, with the variance vanishing in both limits.

For $c > 0$, we derive the equilibrium for this family in Appendix C. Although the equilib-

[25] The density f_β may fail to satisfy the positive density assumption at $\theta = 1$. This does not affect the analysis, since all equilibrium variables are interior.

rium is not available in closed form, it is uniquely pinned down by a single implicit equation.[26]

The linear virtual valuation delivers clean cost comparative statics, because the equilibrium expressions are separable in c: both unit prices and the marginal type increase with c, while both demands and profits decrease. In particular, the cap is independent of c. A higher marginal cost reduces Firm 1's incentive to sell more quantity, but also reduces Firm 2's incentive to cut its price to attract buyers from Firm 1, giving Firm 1 an incentive to raise its cap. For this distribution family, these opposing effects exactly offset.

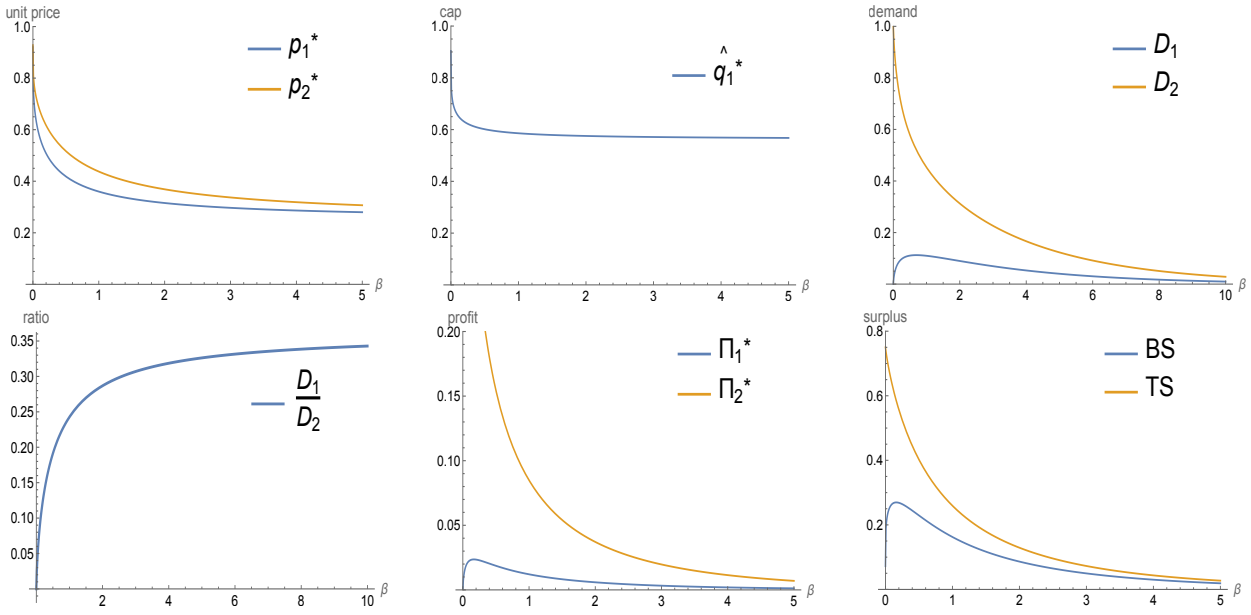


Figure 6: Equilibrium outcomes as β varies when $c = 1/4$.

We next examine how equilibrium outcomes vary with β. Figure 6 illustrates the case $c = 1/4$. As β increases, the distribution shifts toward lower willingness-to-pay types and the upper tail shrinks. Both firms respond by lowering their unit prices, while Firm 1 also tightens its cap. When the upper tail becomes thinner, competing too closely with Firm 2 for high types becomes less valuable; a smaller cap makes Firm 1's offer more differentiated from Firm 2's full-unit offer and helps preserve the segmentation of the market.

Despite this shift toward lower valuations, the effects on demands and profits differ across the two firms. Firm 2 serves the upper segment of the market, so its demand and profit decline monotonically as β increases. Firm 1, however, can initially benefit from the shift. To see why, consider the limit as $\beta \downarrow 0$, where the distribution concentrates at 1. With essentially no heterogeneity to segment, Firm 2's second-mover position allows it to capture almost the entire market and fully extract the buyer's surplus in the limit, while Firm 1's

[26]The condition in Proposition 3 is satisfied if and only if $\beta \geq 1$. Nevertheless, the equilibrium is uniquely characterized for all $\beta > 0$.

demand and profit vanish. Starting from this limit, an increase in β creates heterogeneity and allows Firm 1 to attract more intermediate types with its capped offer, raising its demand and profit. This also makes surplus extraction more difficult for Firm 2, and buyer surplus increases.

As β increases further, Firm 1's demand and profit eventually decline as willingness to pay falls across the market. Nevertheless, D_1/D_2 continues to rise because high-valuation types become scarcer relative to intermediate types. In the limit as $\beta \to \infty$, the distribution concentrates at zero, and both firms' demands and profits, as well as buyer surplus and total surplus, converge to zero.[27]

5 Extensions and Discussion

This section discusses how our analysis may extend beyond the baseline model and highlights several directions for future research.

Asymmetric costs. Our analysis extends to asymmetric marginal costs c_1 and c_2. Let $\delta := c_2 - c_1$ denote the cost difference. When $|\delta|$ is moderate, both firms earn positive profits, and Firm 1's optimal menu still consists of a single quantity-payment option. The resulting equilibrium preserves the same market segmentation: Firm 1 serves intermediate types with a discounted capped offer, while Firm 2 serves high types with its full-unit offer. When δ is sufficiently positive, Firm 1 prefers to deter entry by offering the full unit at a price just below Firm 2's marginal cost. Conversely, when δ is sufficiently negative, Firm 1 may be unable to earn positive profit despite its greater contractual flexibility. Whenever the capped-offer equilibrium exists, entry continues to increase buyer surplus and total surplus relative to Firm 1's monopoly benchmark. Within this equilibrium, industry profit falls when $\delta \geq 0$, but can increase when Firm 2's cost advantage is sufficiently large.

Nonlinear utility. The single-option result continues to hold with convex utility. Suppose that the buyer's payoff is $\theta v(q) - t$, where v is continuous, strictly increasing, and convex, with $v(0) = 0$ and $v(1) = 1$. Convexity preserves inelastic demand under Firm 2's linear pricing: the buyer still chooses either zero or the full unit. Using $v(q)$ as the allocation variable instead of q, the buyer's payoff again has the same linear form as in the baseline model. The single-marginal-type argument therefore continues to apply, and the corresponding set of allocations preserves Firm 2's best response as in Lemma 4. Firm 1's restricted objective is convex: since v^{-1} is concave, production cost is concave in the new allocation, and subtracting

[27] We summarize these comparative statics formally in Lemma C.1.

this cost makes the objective convex. Bauer's Maximum Principle therefore still yields an optimal extreme point, reducing the menu to at most two options. Under regularity, the transformation used in Proposition 2 continues to rule out a two-option menu, so an optimal menu can still be chosen to consist of a single option.[28] The existence and uniqueness of equilibrium, as well as the resulting equilibrium outcomes, generally depend on $v(q)$ and may require additional conditions.

By contrast, allowing for concave utility in quantity is a natural extension, but it would raise several difficulties. The buyer's choice may no longer be summarized by a single marginal type: types may differ not only in which firm they prefer, but also in how much they want to buy from each firm. Thus, preserving Firm 2's best response would generally require a more sophisticated class of Firm 1 allocations, and Firm 1's restricted objective need not be linear or convex in allocation. These differences suggest that the single-option result may no longer hold under concave utility: additional options could become useful not merely for screening within Firm 1's segment, but also for shaping Firm 2's subsequent pricing incentives.

Quality differentiation. Another direction for future research is product-line competition with quality differentiation. While our baseline model assumes that the two firms offer products of the same intrinsic quality, the incumbent may have an established advantage over the entrant in product performance. Suppose, for example, that a type θ buyer obtains utility $v_i q\theta - t$ from consuming quantity $q \in [0,1]$ from firm i, where v_i is firm i's intrinsic quality and $v_1 > v_2$. We continue to represent Firm 1's offer by an allocation-transfer pair (x,t), while Firm 2 is restricted to linear pricing. Firm 1's effective quality for type θ is then $v_1 x(\theta)$, which is endogenous through its allocation even though intrinsic quality is fixed. Such a framework would connect our contract design analysis to product-line strategy.

With quality differentiation, however, the buyer's payoff curves can intersect more than once because Firm 1's effective quality $v_1 x(\theta)$ can exceed Firm 2's quality v_2 for some types. As a result, Firm 2 could serve an intermediate segment, while Firm 1 serves lower and higher types. If multiple intersections can be ruled out for an optimal menu, the logic of our baseline analysis may still be extended to establish the optimality of a single-option menu.

Different equilibrium regimes may arise depending on the quality gap. When the qualities are close, the equilibrium would resemble the baseline market segmentation: Firm 1 offers a lower effective quality option, while high types purchase Firm 2's full-unit option. When Firm 1's quality advantage is large, however, its optimal offer may shift toward a higher effective quality, potentially the full unit, allowing it to retain high types while Firm 2 serves

[28] The single-option result also extends to the baseline model with common increasing, concave production costs $C(q)$, where $C(0) = 0$.

intermediate types at a lower price.

Multi-sourcing. Allowing the buyer to purchase from both firms is also an interesting extension, but it raises a different challenge. Even with linear utility, high types may split their purchases between the two firms. In this case, Firm 2's price affects which option the buyer selects from Firm 1's menu and hence the quantity purchased from each firm. Although Firm 1's objective remains linear when Firm 2's best response is fixed, preserving that response imposes different constraints on Firm 1's allocation. Consequently, the extreme-point argument no longer yields the same restriction on menu complexity, and menus with more than two options can be optimal.

Other extensions. Two further directions are to allow the entrant to offer more flexible contracts in our sequential setting and to study symmetric simultaneous menu competition. In the first extension, the question is whether the entrant would still find it optimal to use the simplest instrument, namely linear pricing. Our intuition is that this simple instrument may remain optimal for the second mover, and it would be interesting to characterize when this is the case. In the second extension, both firms would compete simultaneously in menus. This would clarify whether richer contracting on both sides drives the market toward a Bertrand-like outcome, such as marginal cost pricing, or whether firms can still sustain differentiated offers for different buyer types.

6 Conclusion

This paper studies optimal contract design under entry. We consider a sequential duopoly in which an incumbent first offers an arbitrary quantity-payment menu to a privately informed buyer, while an entrant subsequently uses linear pricing. Although the incumbent has a richer contract space, we show that it can optimally use a single quantity-payment option. Under inelastic demand, this option can be implemented as linear pricing with a quantity cap.

The capped offer leads to competitive market segmentation. In equilibrium, the incumbent charges a lower unit price but limits the quantity available under its offer. This allows the incumbent to serve intermediate types without competing directly for the high types, who instead buy the entrant's full-unit product. Thus the incumbent does not use its ability to offer arbitrary menus to screen all buyer types. Instead, it uses a limited version of its product to shape the entrant's residual demand, turning screening into segmentation across firms.

The equilibrium also has clear profit and welfare implications. The entrant earns more than the incumbent because it captures the upper segment at a higher unit price. Thus, the capped offer allows the incumbent to retain part of the market, but it cannot eliminate the entrant's advantage from moving second. Relative to the monopoly benchmark, entry expands trade to lower-value types and lowers equilibrium prices. As a result, surplus shifts away from firms and toward the buyer: industry profit falls, while buyer surplus and total surplus increase.

Appendix A Proofs

Proof of Lemma 1. (Proof of the $\mathcal{C} \Rightarrow \mathcal{M}$ direction) Fix $C_T \in \mathcal{C}$ and let

$$u_T(\theta) = \max_{(q,t)\in C_T} \{\theta q - t\},\ \theta \in \Theta.$$

Since $(0,0) \in C_T$, we have $u_T(\theta) \geq 0$ for all θ.

Set $x(\theta) := q_T(\theta)$ and $t(\theta) := t_T(\theta)$ for every $\theta \in \Theta$. Since $(q_T(\theta), t_T(\theta))$ maximizes type θ's utility over C_T, for all θ, θ',

$$\theta x(\theta) - t(\theta) \geq \theta x(\theta') - t(\theta'),$$

so (IC) holds. Moreover, $\theta x(\theta) - t(\theta) = u_T(\theta) \geq 0$, so (IR) is satisfied. Therefore, the constructed $(x,t) \in \mathcal{M}$ satisfies $(q_T(\theta), t_T(\theta)) = (x(\theta), t(\theta))$ for every $\theta \in \Theta$, and $u_T(\theta) = \theta x(\theta) - t(\theta)$ for every $\theta \in \Theta$.

(Proof of the $\mathcal{M} \Rightarrow \mathcal{C}$ direction) Fix $(x,t) \in \mathcal{M}$ and let $u_M(\theta) = \theta x(\theta) - t(\theta)$. By (IC), u_M is convex and non-decreasing, and $x(\theta) \in \partial u_M(\theta)$ for every θ. For $q \in [0,1]$, define

$$T(q) := u_M^*(q) = \sup_{v\in\Theta} \{vq - u_M(v)\},$$

where u_M^* is the convex conjugate of u_M. As a supremum of continuous affine functions of q, T is lower semicontinuous. Hence $T \in \mathcal{T}$. Let

$$C_T = \{(q, T(q)) : q \in [0,1]\} \cup \{(0,0)\}.$$

By the Fenchel–Young inequality, $u_M(\theta) + T(q) \geq \theta q$ for every $q \in [0,1]$. Since $x(\theta) \in \partial u_M(\theta)$, equality holds at $q = x(\theta)$. Hence

$$\max_{q\in[0,1]} \{\theta q - T(q)\} = u_M(\theta)$$

for every θ. Since $u_M(\theta) \geq 0$, adding the outside option $(0,0)$ does not change the maximum. It follows that $u_T(\theta) = u_M(\theta)$ for every θ.

By definition, $(q_T(\theta), t_T(\theta))$ maximizes type θ's utility over C_T. Hence, for all θ, θ', $u_M(\theta') = u_T(\theta') \geq \theta' q_T(\theta) - t_T(\theta) = u_M(\theta) + (\theta' - \theta) q_T(\theta)$, so $q_T(\theta) \in \partial u_M(\theta)$. Since u_M is convex, $\partial u_M(\theta)$ is a singleton almost everywhere. Because $x(\theta) \in \partial u_M(\theta)$ as well, it follows that $q_T(\theta) = x(\theta)$ almost everywhere. Moreover, $t_T(\theta) = \theta q_T(\theta) - u_T(\theta) = \theta x(\theta) - u_M(\theta) = t(\theta)$ almost everywhere.

Therefore, the constructed $C_T \in \mathcal{C}$ satisfies $(q_T(\theta), t_T(\theta)) = (x(\theta), t(\theta))$ almost everywhere, while $u_T(\theta) = \theta x(\theta) - t(\theta)$ for every $\theta \in \Theta$. Note that adding the outside option $(0,0)$ in C_T does not change allocations and transfers. When $T(0) = 0$, the menu already includes the outside option. When $T(0) < 0$, $(0,0)$ is never chosen by any type since it is strictly dominated by $(0, T(0))$.[29] □

Proof of Lemma 2. Fix an allocation x offered by Firm 1 and a best response price $p_2^\star$ by Firm 2. If Firm 1's profit is non-positive, the allocation $x' \equiv 0$ yields zero profit, so the conclusion follows. Suppose that Firm 1's profit is positive. If $x(\theta) < 1$ almost everywhere, the conclusion follows by taking $x' = x$, since Lemma 3 implies that Firm 2's best-response marginal type is unique and hence Firm 1 obtains the same profit.

We first show that Firm 2 also earns positive profit. Suppose, to the contrary, that its profit is zero. Since Firm 2 best responds to Firm 1's offer, no price $p_2 > c$ can yield positive profit. This requires $\int_0^\theta x(s)ds \geq \max\{0, \theta - c\}$, $\forall \theta \in \Theta$. Indeed, if this inequality does not hold for some type, Firm 2 can choose a price slightly above c and attract a positive measure of types at a positive margin.

Using $t(\theta) = \theta x(\theta) - \int_0^\theta x(s)ds$, Firm 1's profit from each type θ it serves is $(\theta - c)x(\theta) - \int_0^\theta x(s)ds$. Under the condition above, this expression is always non-positive, since $(\theta - c)x(\theta) - \int_0^\theta x(s)ds \leq (\theta - c)x(\theta) - \max\{\theta - c, 0\} \leq 0$. Hence, if Firm 2 earns zero profit, Firm 1 cannot earn positive profit either, a contradiction. Thus Firm 2 earns positive profit.

Now suppose that $x(\theta) = 1$ on a set of positive measure. Since x is non-decreasing, there exists $\eta \in [0,1)$ such that $x(\theta) < 1$ for $\theta < \eta$ and $x(\theta) = 1$ for $\theta > \eta$. Hence, $\int_0^\theta x(s)ds = \theta - k_1$ for all $\theta \in [\eta, 1]$, for some constant k_1. Define $U(\theta | x, p_2) := \theta - p_2 - \int_0^\theta x(s)ds$. Because $x(\theta) < 1$ for $\theta < \eta$, $U(\cdot | x, p_2^\star)$ is strictly increasing on $[0, \eta]$. Since $x(\theta) = 1$ for $\theta > \eta$, it is constant on $[\eta, 1]$ and equal to $k_1 - p_2^\star$.

We must have $p_2^\star \leq k_1$. Otherwise, $U(\theta | x, p_2^\star) < 0$ for every θ, so Firm 2 would face zero demand, contradicting the fact that it earns positive profit. Hence there is a unique

[29] $T(0)$ cannot be positive by the definition of $T(q)$.

$z^* \in (0, \eta]$ such that

$$U(z^*|x, p_2^*) = 0 \Leftrightarrow p_2^* = \int_0^{z^*} [1 - x(s)]ds.$$

If $p_2^* < k_1$, then $z^* < \eta$. If $p_2^* = k_1$, then $z^* = \eta$, and all types above η purchase from Firm 2 under the tie-breaking rule.

We next show that $x(z^*-) < 1$. This follows immediately if $z^* < \eta$. Suppose that $z^* = \eta$, so that $p_2^* = k_1$. For every $z \in [0, \eta]$, define the price $p(z|x) := \int_0^z [1 - x(s)]ds$. Since $x(\theta) < 1$ for $\theta < \eta$, the mapping $z \mapsto p(z|x)$ is strictly increasing on $[0, \eta]$, with $p(\eta|x) = k_1 = p_2^*$. Thus, because p_2^* is a best response, η maximizes

$$\big(1 - F(z)\big)\Big(\int_0^z [1 - x(s)]ds - c\Big)$$

over $z \in [0, \eta]$. Its left derivative at η must therefore be non-negative:

$$-f(\eta)(k_1 - c) + (1 - F(\eta))\big(1 - x(\eta-)\big) \geq 0.$$

Since $k_1 = p_2^* > c$, it follows that $x(\eta-) < 1$.

Consider the allocation $x'(\theta) = x(\theta)$ for $\theta < z^*$ and $x'(\theta) = x(z^*-)$ for $\theta \geq z^*$. By construction, $x' \in \mathcal{X}$ and $x'(\theta) < 1$ almost everywhere. Then each price of Firm 2 induces a unique marginal type, and Firm 2's profit is continuous in the marginal type, so a best response to x' exists. By Lemma 3 below, its best-response marginal type z' is unique. Moreover, $z' \geq z^*$. This is because x' and x coincide below z^* and Firm 2's profit is unchanged below z^*.

Firm 1's profit from types below z^* is unchanged. Moreover, the net profit $(\theta - c)x(\theta) - \int_0^\theta x(s)ds$ is non-decreasing for $\theta \geq c$, by monotonicity of x. Since Firm 1's profit under x and p_2^* is positive, it follows that $(z^* - c)x(z^*-) - \int_0^{z^*} x(s)ds \geq 0$. Otherwise, every type purchasing from Firm 1 would generate strictly negative profit. Under x', each additional type in (z^*, z') receives quantity $x(z^*-)$ and generates this same non-negative profit. Hence, Firm 1's profit under x' and any best response of Firm 2 is weakly higher than its profit under x and p_2^*. □

Proof of Lemma 3. We show that $g(\cdot|x)$ is strictly increasing on $[0, 1)$. Observe that for any $0 \leq \alpha_1 < \alpha_2 < 1$,

$$\begin{aligned} &g(\alpha_2|x) - g(\alpha_1|x) \\ &= \int_{\alpha_1}^{\alpha_2} [1 - x(s)]ds - \frac{1 - F(\alpha_2)}{f(\alpha_2)}\big(1 - x(\alpha_2)\big) + \frac{1 - F(\alpha_1)}{f(\alpha_1)}\big(1 - x(\alpha_1)\big) \end{aligned}$$

$$
\begin{aligned}
&\geq (\alpha_2 - \alpha_1)\big(1 - x(\alpha_2)\big) - \left(\frac{1 - F(\alpha_2)}{f(\alpha_2)} - \frac{1 - F(\alpha_1)}{f(\alpha_1)}\right)\big(1 - x(\alpha_2)\big) \\
&= \big(1 - x(\alpha_2)\big)\big(J(\alpha_2) - J(\alpha_1)\big) \\
&> 0
\end{aligned}
$$

where in the first inequality we use the fact that $\int_0^\theta [1 - x(s)]ds$ is concave in θ and thus $\int_{\alpha_1}^{\alpha_2} [1 - x(s)]ds \geq (\alpha_2 - \alpha_1)(1 - x(\alpha_2))$, together with $1 - x(\alpha_1) \geq 1 - x(\alpha_2)$. The second inequality follows from the regularity of F and $1 - x(\alpha_2) > 0$, since $\alpha_2 < 1$ and $x(\theta) < 1$ almost everywhere.

Since $(1 - F(z))/f(z) \to 0$ as $z \uparrow 1$, we also have $g(1|x) = \lim_{z \uparrow 1} g(z|x)$. Hence, $g(\cdot|x)$ is strictly increasing on Θ. □

Proof of Lemma 4. First, note that $z_x > 0$, since $g(0 + |x) = -c - (1 - x(0+))/f(0) < 0$. If $z_x = 1$, then optimality of z_x under x implies $g(1 - |x) \leq 0$. For any $\tilde{x} \in \Delta_x$, since $\int_0^1 \tilde{x}(s)ds = \int_0^1 x(s)ds$, we have $g(1 - |\tilde{x}) = g(1 - |x) \leq 0$. Therefore, $z_{\tilde{x}} = 1 = z_x$.

It remains to consider $z_x \in (0, 1)$. It is enough to show that $g(z_x + |\tilde{x}) \geq 0$ and $g(z_x - |\tilde{x}) \leq 0$ for any $\tilde{x} \in \Delta_x$. Since z_x is Firm 2's best response given x and x is continuous at z_x, it must be the case that $g(z_x + |x) = g(z_x - |x) = 0$. Using $\int_0^{z_x} \tilde{x}(s)ds = \int_0^{z_x} x(s)ds$ for any $\tilde{x} \in \Delta_x$,

$$
g(z_x + |\tilde{x}) - g(z_x + |x) = \frac{1 - F(z_x)}{f(z_x)}\big(\tilde{x}(z_x+) - x(z_x+)\big) = \frac{1 - F(z_x)}{f(z_x)}\big(x(z_x) - x(z_x)\big) = 0,
$$

where the second equality holds since $\tilde{x}(\theta) = x(z_x)$ for $\theta \in (z_x, 1]$. Hence, $g(z_x + |\tilde{x}) = g(z_x + |x) = 0$ follows. To show $g(z_x - |\tilde{x}) \leq 0$, note that

$$
g(z_x - |x) - g(z_x - |\tilde{x}) = \frac{1 - F(z_x)}{f(z_x)}\big(x(z_x-) - \tilde{x}(z_x-)\big) \geq \frac{1 - F(z_x)}{f(z_x)}\big(x(z_x) - \tilde{x}(z_x+)\big) = 0,
$$

where the inequality uses monotonicity of $\tilde{x}$. Hence, $g(z_x - |\tilde{x}) \leq g(z_x - |x) = 0$.

Finally, we show that it is without loss to focus on allocations that are continuous at the induced best response.

Lemma A.1. *Fix $x \in \mathcal{X}$, and let z_x denote Firm 2's best response induced by x. If x is discontinuous at z_x, then there exist an allocation $x' \in \mathcal{X}$ and the corresponding best response $z_{x'}$ such that x' is continuous at $z_{x'}$, and Firm 1's profit under x' is weakly higher than under x.*

Proof. We show that there is a weakly profitable deviation by Firm 1 if x is discontinuous at z_x. Consider a deviation by offering x' such that $x'(\theta) = x(\theta)$ for $\theta \in [0, z_x)$ and $x'(\theta) = x(z_x-)$ for $\theta \in [z_x, 1]$. Then $g(z|x') = g(z|x)$ for $z < z_x$, and thus $\pi_2(z|x) = \pi_2(z|x')$ for $z \leq z_x$. Let $z_{x'}$

denote Firm 2's best response under x'. Then $z_{x'} \geq z_x$, because Firm 2's profit is unchanged over $z \in [0, z_x]$. Since x' is continuous at z_x and constant on $[z_x, 1]$, it follows that x' is continuous at $z_{x'}$. If Firm 1's profit under x is negative, the zero allocation is continuous and yields weakly higher profit. Hence, suppose without loss that Firm 1's profit under x is non-negative. Firm 1's profit from types in $[0, z_x]$ is also unchanged. Moreover, by the same argument as in Lemma 2, every additional type served under x' generates non-negative profit. Therefore, Firm 1 is weakly better off under x'. □

□

Proof of Lemma 6. We have $\int_0^{z_x} w(s)ds = (z_x - \lambda)x(z_x) = \int_0^{z_x} x(s)ds$. Since it is clear from the definition that $MPS(w) \subset \Delta_x^\circ$, it suffices to prove $\Delta_x^\circ \subset MPS(w)$. To show this, fix any $\tilde{x}^\circ \in \Delta_x^\circ$. By definition, $\int_0^{z_x} \tilde{x}^\circ(s)ds = \int_0^{z_x} x(s)ds = \int_0^{z_x} w(s)ds$ and $0 \leq \tilde{x}^\circ(\theta) \leq x(z_x)$ for all $\theta \in [0, z_x]$. Then, for any $\theta \in [0, z_x]$,

$$\int_0^\theta \tilde{x}^\circ(s)\,ds = \int_0^{z_x} \tilde{x}^\circ(s)\,ds - \int_\theta^{z_x} \tilde{x}^\circ(s)\,ds \geq \int_0^{z_x} w(s)\,ds - (z_x - \theta)\,x(z_x).$$

Using $(z_x - \lambda)x(z_x) = \int_0^{z_x} w(s)ds$, the lower bound becomes $(z_x - \lambda)\,x(z_x) - (z_x - \theta)\,x(z_x) = (\theta - \lambda)\,x(z_x)$. Since

$$\int_0^\theta w(s)\,ds = \begin{cases} 0 & \text{if } \theta < \lambda, \\ (\theta - \lambda)\,x(z_x) & \text{if } \theta \geq \lambda, \end{cases}$$

we have $\int_0^\theta \tilde{x}^\circ(s)\,ds \geq \int_0^\theta w(s)\,ds$ for all $\theta \in [0, z_x]$. Therefore $\tilde{x}^\circ \in MPS(w)$, proving $\Delta_x^\circ \subset MPS(w)$. □

Proof of Proposition 1. We establish two results omitted from the main text: (i) Δ_x° is convex and compact, and (ii) the structure of the extreme points of Δ_x°.

To show (i), we endow Δ_x° with the $L^1([0, z_x])$ topology. Convexity is immediate. To establish compactness, let $(y_n) \subset \Delta_x^\circ$. Since the functions y_n are non-decreasing and uniformly bounded between 0 and $x(z_x)$, Helly's selection theorem yields a subsequence (y_{n_k}) and a non-decreasing function $y : [0, z_x] \to [0, x(z_x)]$ such that $y_{n_k}(\theta) \to y(\theta)$ at every continuity point of y. Since a monotone function has at most countably many discontinuities, this convergence holds almost everywhere. By the dominated convergence theorem, $y_{n_k} \to y$ in L^1, and

$$\int_0^{z_x} y(s)\,ds = \lim_{k\to\infty} \int_0^{z_x} y_{n_k}(s)\,ds = \int_0^{z_x} x(s)\,ds.$$

Thus $y \in \Delta_x^\circ$. Hence Δ_x° is compact in $L^1([0, z_x])$.

Next, we claim that the extreme points of Δ_x° have a particularly simple structure. Since $x(z_x) \cdot \mathbb{1}\{\theta \in [\lambda, z_x]\}$ is a step function, Lemma 5 implies that the extreme points

of $\text{MPS}\big(x(z_x)\cdot\mathbb{1}\{\theta\in[\lambda,z_x]\}\big)$ can be classified according to whether λ is contained in a pooling interval. Hence any extreme point $\hat{x}^\circ$ must coincide almost everywhere with one of the following functions:[30]

$$\hat{x}^\circ(\theta)=\begin{cases}0 & \text{if } \theta\in[0,\theta_1]\\ \frac{(z_x-\lambda)x(z_x)}{z_x-\theta_1} & \text{if } \theta\in(\theta_1,z_x]\end{cases},$$

$$\text{or } \hat{x}^\circ(\theta)=\begin{cases}\frac{(\theta_2-\lambda)x(z_x)}{\theta_2} & \text{if } \theta\in[0,\theta_2]\\ x(z_x) & \text{if } \theta\in(\theta_2,z_x]\end{cases}, \text{ or } \hat{x}^\circ(\theta)=\begin{cases}0 & \text{if } \theta\in[0,\theta_3]\\ \frac{(\theta_4-\lambda)x(z_x)}{\theta_4-\theta_3} & \text{if } \theta\in(\theta_3,\theta_4]\\ x(z_x) & \text{if } \theta\in(\theta_4,z_x]\end{cases}$$

for some $\theta_1,\theta_2,\theta_3$, and θ_4 such that $0<\theta_1\le\lambda$, $\lambda\le\theta_2<z_x$, and $0<\theta_3<\lambda<\theta_4<z_x$. Therefore, the extension $\hat{x}\in\Delta_x$ of an optimal extreme point $\hat{x}^\circ$ solves (7) and consists of at most two options. Each of these forms is illustrated in Figure 3. □

Proof of Proposition 2. Consider the transformation described in the text, which converts any two-option allocation $\hat{x}\in\Delta_x$ obtained in Proposition 1 into an allocation $x^{(1)}$ with a single option. Suppose without loss of generality that Firm 1's profit under $\hat{x}$ is positive. We show that this transformation strictly increases Firm 1's profit.

By Lemma 4, together with $a_2\le z_x$, the best response of Firm 2 does not change. Note that the payments of each type, under $\hat{x}$ and $x^{(1)}$ respectively, are $\hat{t}(\theta)=\theta\hat{x}(\theta)-\int_0^\theta\hat{x}(s)ds$ and $t^{(1)}(\theta)=\theta x^{(1)}(\theta)-\int_0^\theta x^{(1)}(s)ds$. Since $\hat{x}(\theta)=x^{(1)}(\theta)$ and $\hat{t}(\theta)=t^{(1)}(\theta)$ for $\theta\in[0,a_1]\cup(a_2,z_x]$, it is enough to consider the payments of types in $(a_1,a_2]$. Let

$$\hat{T}:=\int_{a_1}^{a_2}[\hat{t}(\theta)-c\hat{x}(\theta)]dF(\theta)=\big(F(a_2)-F(a_1)\big)b_2(a_1-c),$$
$$T^{(1)}:=\int_{a_1}^{a_2}[t^{(1)}(\theta)-cx^{(1)}(\theta)]dF(\theta)=b_3(\tilde{a}-c)\big(F(a_2)-F(\tilde{a})\big).$$

To prove the result, we will show

$$T^{(1)}-\hat{T}>0. \tag{13}$$

First, we show that $a_2>c$. Suppose, to the contrary, that $a_2\le c$. Since $b_1=0$, for every $\theta\in(a_1,a_2]$, $\hat{t}(\theta)-c\hat{x}(\theta)=b_2(a_1-c)<0$, and for every $\theta\in(a_2,z_x]$, $\hat{t}(\theta)-c\hat{x}(\theta)=b_2(a_1-c)+(b_3-b_2)(a_2-c)<0$. Thus Firm 1's profit under $\hat{x}$ is strictly negative, contradicting the assumption that it is positive. Hence, $a_2>c$.

[30] We rule out the case when $\hat{x}^\circ$ is constant because it generates non-positive profit for Firm 1 and thus it is never optimal.

Note that

$$T^{(1)} - \hat{T} = (a_2 - c)(b_3 - b_2)\big(F(a_2) - F(\tilde{a})\big) - (a_1 - c)b_2\big(F(\tilde{a}) - F(a_1)\big).$$

If $a_1 \leq c$, then the expression above is strictly positive. In particular, this covers the case $a_1 = 0$. Hence, we assume $a_1 > c$. Then (13) becomes

$$\begin{aligned} T^{(1)} - \hat{T} > 0 &\Leftrightarrow (a_2 - c)(b_3 - b_2)\big(F(a_2) - F(\tilde{a})\big) > (a_1 - c)b_2\big(F(\tilde{a}) - F(a_1)\big) \\ &\Leftrightarrow \frac{F(a_2) - F(\tilde{a})}{a_2 - \tilde{a}}(a_2 - c) > \frac{F(\tilde{a}) - F(a_1)}{\tilde{a} - a_1}(a_1 - c) \end{aligned} \tag{14}$$

where we use $b_2/(b_3 - b_2) = (a_2 - \tilde{a})/(\tilde{a} - a_1)$ in the second line. The following lemma provides a sufficient condition for (14) to hold.

Lemma A.2. *(McAfee and McMillan (1987)) If F is regular, then $\big(1/(1 - F(\theta))\big)'' > 0$ for all $\theta \in (0, 1)$.*

Proof. The condition for F being regular is

$$\left(\theta - \frac{1 - F(\theta)}{f(\theta)}\right)' > 0 \Leftrightarrow 2f(\theta)^2 + \big(1 - F(\theta)\big)f'(\theta) > 0. \tag{15}$$

Meanwhile,

$$\left(\frac{1}{1 - F(\theta)}\right)'' > 0 \Leftrightarrow \big(1 - F(\theta)\big)^{-3}\big(2f(\theta)^2 + (1 - F(\theta))f'(\theta)\big) > 0. \tag{16}$$

It is clear that (15) implies (16). □

By Lemma A.2, if F is regular, $\big(1 - F\big)^{-1}$ is strictly convex, implying

$$\begin{aligned} &\frac{\big(1 - F(\tilde{a})\big)^{-1} - \big(1 - F(a_1)\big)^{-1}}{\tilde{a} - a_1} < \frac{\big(1 - F(a_2)\big)^{-1} - \big(1 - F(\tilde{a})\big)^{-1}}{a_2 - \tilde{a}} \\ &\Leftrightarrow \frac{F(\tilde{a}) - F(a_1)}{\tilde{a} - a_1} < \frac{F(a_2) - F(\tilde{a})}{a_2 - \tilde{a}}\frac{1 - F(a_1)}{1 - F(a_2)}. \end{aligned}$$

Hence, to show that (14) holds, it suffices to show that

$$\frac{a_1 - c}{a_2 - c} \leq \frac{1 - F(a_2)}{1 - F(a_1)}. \tag{17}$$

z_x is assumed to be in $[a_2, 1]$ and thus $\pi_2(z|\hat{x})$ is strictly increasing over $[a_1, a_2]$. Hence, we have $\pi_2(a_1|\hat{x}) < \pi_2(a_2|\hat{x}) \Leftrightarrow (a_1 - c)\big(1 - F(a_1)\big) < \big(a_2 - (a_2 - a_1)b_2 - c\big)\big(1 - F(a_2)\big)$. Since $a_1 > c$

and $0 < b_2 < 1$, $a_2 - (a_2 - a_1)b_2 = b_2 a_1 + (1 - b_2)a_2$ is a convex combination of a_1 and a_2, and hence is greater than c. Therefore, we have

$$\frac{1 - F(a_2)}{1 - F(a_1)} > \frac{a_1 - c}{a_2 - (a_2 - a_1)b_2 - c} > \frac{a_1 - c}{a_2 - c},$$

where the second inequality follows from $a_2 - (a_2 - a_1)b_2 - c < a_2 - c$. This proves (17), and hence (14) holds. Therefore, $T^{(1)} - \hat{T} > 0$.

Now we show that every maximizer of (5) has a single option below its induced marginal type. Since Firm 1 can earn strictly positive profit, every maximizer earns positive profit. First, consider a maximizer x satisfying $x(\theta) < 1$ almost everywhere and continuous at z_x. By Lemma 4, every element of Δ_x°, extended to Δ_x, induces the same marginal type z_x. Since x is globally optimal, its truncation $x|_{[0,z_x]}$ maximizes (8).

Let $\mathcal{E} \subseteq \Delta_x^\circ$ denote the set of maximizers of (8). Since Δ_x° is compact and convex and the objective is continuous and linear, $\mathcal{E}$ is a nonempty compact convex face of Δ_x° in $L^1([0, z_x])$. Consider an extreme point $\hat{x}^\circ$ of $\mathcal{E}$, which is also an extreme point of Δ_x°, and its extension $\hat{x} \in \Delta_x$. This extension earns the same positive profit as x. By the extreme-point characterization in the proof of Proposition 1, either $\hat{x}^\circ = w$ almost everywhere, where w is the step function defined in the text, or $\hat{x}$ has two options. The latter case is ruled out by the preceding argument. Thus w is the only extreme point of $\mathcal{E}$. Applying Bauer's Maximum Principle to the continuous convex function $\sigma \mapsto \|\sigma - w\|_{L^1([0,z_x])}$ on $\mathcal{E}$, its maximum is attained at w and therefore equals zero. Hence $\mathcal{E} = \{w\}$, so $x = w$ almost everywhere on $[0, z_x]$.

Finally, consider a maximizer x that fails either of the initial conditions. The constructions in the proofs of Lemma 2 and Lemma A.1 yield an allocation x' satisfying $x'(\theta) < 1$ almost everywhere and continuous at $z_{x'}$, with profit weakly higher than that of x. Since x is globally optimal, x' is also a maximizer. The preceding argument therefore implies that x' has a single option below $z_{x'}$. Moreover, these constructions give $x' = x$ on $[0, z_x)$ and $z_{x'} \geq z_x$. Thus x has at most one positive option below z_x. Since x earns positive profit, it must have exactly one. □

Proof of Theorem 1. We prove existence by first considering a relaxation of Firm 1's problem and then showing that any maximizer of the relaxed problem is interior. Define

$$K_c := \{(p_1, \hat{q}_1, z) \in [c, 1] \times [0, 1] \times [c, 1] : p_1 \leq z,\ (1 - \hat{q}_1)J(z) + \hat{q}_1 p_1 = c\}.$$

The set K_c is compact and the objective $\hat{q}_1(p_1 - c)\big[F(z) - F(p_1)\big]$ is continuous. Hence the relaxed problem

$$\max_{(p_1, \hat{q}_1, z) \in K_c} \hat{q}_1(p_1 - c)\big[F(z) - F(p_1)\big]$$

admits a maximizer.

The maximum value of the relaxed problem is strictly positive. Since $J(c) = c - \big(1 - F(c)\big)/f(c) < c$, we can choose $\varepsilon > 0$ small enough that $z_\varepsilon := c + \varepsilon < 1$ and $J(z_\varepsilon) < c$. Set $p_{1,\varepsilon} := c + \varepsilon/2$ and

$$\hat{q}_{1,\varepsilon} := \frac{c - J(z_\varepsilon)}{p_{1,\varepsilon} - J(z_\varepsilon)}.$$

Then $0 < \hat{q}_{1,\varepsilon} < 1$, $c < p_{1,\varepsilon} < z_\varepsilon < 1$, and $(p_{1,\varepsilon}, \hat{q}_{1,\varepsilon}, z_\varepsilon) \in K_c$. Moreover, $\hat{q}_{1,\varepsilon}(p_{1,\varepsilon} - c)\big[F(z_\varepsilon) - F(p_{1,\varepsilon})\big] > 0$. Thus the relaxed problem has a strictly positive value.

Let $(p_1^*, \hat{q}_1^*, z^*)$ be a maximizer of the relaxed problem. The preceding arguments rule out the zero-profit boundaries. If $\hat{q}_1^* = 0$, the objective is zero. If $\hat{q}_1^* = 1$, the constraint gives $p_1^* = c$, so the objective is again zero. Hence $0 < \hat{q}_1^* < 1$. If $p_1^* = c$, the objective is zero, so $p_1^* > c$. If $p_1^* = z^*$, then $F(z^*) - F(p_1^*) = 0$, so the objective is zero. Hence $p_1^* < z^*$.

Finally, $z^* < 1$. Suppose instead that $z^* = 1$. Since $J(1) = 1$, the constraint gives $1 - \hat{q}_1^* + \hat{q}_1^* p_1^* = c$. But $0 < \hat{q}_1^* < 1$ and $p_1^* > c$, so the left-hand side is a strict convex combination of 1 and a number strictly larger than c. It is therefore strictly larger than c, a contradiction. Thus $0 < \hat{q}_1^* < 1$ and $c < p_1^* < z^* < 1$. Therefore the maximizer of the relaxed problem is feasible for the original problem (11), and it solves (11).

Given this solution, define $p_2^* := (1 - \hat{q}_1^*)z^* + \hat{q}_1^* p_1^*$. Using the constraint,

$$p_2^* - c = (1 - \hat{q}_1^*)\big(z^* - J(z^*)\big) = (1 - \hat{q}_1^*)\frac{1 - F(z^*)}{f(z^*)}.$$

Thus the first order condition for Firm 2's problem holds at z^*. By Lemma 3, p_2^* is Firm 2's best response to Firm 1's capped option. The market segmentation follows from the buyer's payoff comparison: types in $[z^*, 1]$ purchase from Firm 2, types in (p_1^*, z^*) purchase from Firm 1, and types in $[0, p_1^*]$ opt out. □

Proof of Proposition 3. By Theorem 1, a maximizer exists and every maximizer is interior. We establish uniqueness by first optimizing Firm 1's price for a fixed cap and then optimizing over the cap.

Fix $\hat{q}_1 \in (0, 1)$. The constraint (10) uniquely determines $z(p_1, \hat{q}_1)$ with $z_{p_1} = -\hat{q}_1/[(1 - \hat{q}_1)J'(z)]$. At any feasible interior choice, $c < p_1 < z$, and (10) implies $J(z) < c$. Hence $p_1, z \in (c, J^{-1}(c))$. Firm 1's first order condition with respect to p_1 is

$$F(z) - F(p_1) = (p_1 - c)f(p_1) + \big(c - J(z)\big)\frac{f(z)}{J'(z)}. \tag{18}$$

We first show that Firm 1's profit is strictly concave in p_1 for any fixed cap. For $p \in$

$(c, J^{-1}(c))$, regularity implies $J(p) < c$. Note that

$$-\pi^m_{pp}(p;c) = \frac{f(p)}{p - J(p)}\big[(p-c)J'(p) + 2(c - J(p))\big] > 0.$$

Moreover, the condition in Proposition 3 implies that $f(z)/J'(z)$ is weakly decreasing in z. Differentiating Firm 1's profit twice with respect to p_1, taking into account Firm 2's response, gives

$$\frac{\partial^2 \pi_1}{\partial p_1^2} = \hat{q}_1 \left[\pi^m_{pp}(p_1;c) - \frac{\hat{q}_1}{(1-\hat{q}_1)J'(z)} \left(2f(z) - \big(c - J(z)\big)\frac{d}{dz}\frac{f(z)}{J'(z)} \right) \right] < 0.$$

Since Firm 1's profit is zero at the boundaries $p_1 = c$ and $p_1 = z$ and strictly positive in between, there is a unique interior optimal price $p_1^*(\hat{q}_1)$ for each cap.

We next show that both $p_1^*(\hat{q}_1)$ and the induced marginal type $z(p_1^*(\hat{q}_1), \hat{q}_1)$ are strictly decreasing in the cap. In (18), the difference between the left-hand side and the right-hand side is strictly decreasing in p_1 holding z fixed, with derivative $\pi^m_{pp}(p_1;c) < 0$, and strictly increasing in z holding p_1 fixed, with derivative

$$2f(z) - \big(c - J(z)\big)\frac{d}{dz}\frac{f(z)}{J'(z)} > 0.$$

Therefore, along the first-order condition (18), p_1 is strictly increasing in z. By (10),

$$\frac{\hat{q}_1}{1-\hat{q}_1} = \frac{c - J(z)}{p_1 - c}.$$

The right-hand side is strictly decreasing in z along the first-order condition, since $J(z)$ and p_1 are both strictly increasing in z. It follows that $p_1^*(\hat{q}_1)$ and $z(p_1^*(\hat{q}_1), \hat{q}_1)$ are both strictly decreasing in $\hat{q}_1$.

We now optimize over the cap. By the envelope theorem, using (10) and (18), we obtain

$$\frac{d}{d\hat{q}_1}\pi_1(p_1^*(\hat{q}_1), \hat{q}_1) = (p_1^* - c)^2 f(p_1^*) - \big(c - J(z)\big)^2 \frac{f(z)}{J'(z)}, \tag{19}$$

where $p_1^* = p_1^*(\hat{q}_1)$ and $z = z(p_1^*(\hat{q}_1), \hat{q}_1)$. We show that the right-hand side of (19) is strictly decreasing in $\hat{q}_1$. First,

$$\frac{d}{dp}\big[(p-c)^2 f(p)\big] = -(p-c)\pi^m_{pp}(p;c) > 0.$$

Since $p_1^*(\hat{q}_1)$ is strictly decreasing, the first term in (19) is strictly decreasing in $\hat{q}_1$. Second,

$(c - J(z))^2$ is strictly decreasing in z, while $f(z)/J'(z)$ is weakly decreasing by assumption. Hence their product is strictly decreasing in z. Since $z(p_1^*(\hat{q}_1), \hat{q}_1)$ is strictly decreasing in $\hat{q}_1$, the second term in (19) is strictly increasing in the cap.

Therefore, Firm 1's profit evaluated at its optimal price is strictly concave in $\hat{q}_1$. Since a maximizer exists and is interior, the optimal cap is unique. The uniqueness of $p_1^*(\hat{q}_1)$ then pins down a unique optimal price, and (10) determines a unique marginal type z. Thus (11) has a unique solution.

By Proposition 2, every optimal allocation has a single option below its induced marginal type. Extending this option to all types can only weakly increase the marginal type, since Firm 2's profit remains unchanged below the original marginal type. A strict increase would contradict Firm 1's optimality. Hence the uniqueness of the solution to (11) establishes the claim. □

Proof of Proposition 4. Part (i). The monopoly first-order condition gives $J(p^m) = c$. By (10), we have $J(z^*) < c < p_1^*$. Thus $J(z^*) < J(p^m)$, and regularity of F implies $z^* < p^m$. Moreover, since p_2^* is a strict convex combination of p_1^* and z^*, we have $p_1^* < p_2^* < z^*$. Hence $p_1^* < p_2^* < z^* < p^m$.

Part (ii). The equilibrium profits are

$$\Pi_1^* = \hat{q}_1^*(p_1^* - c)\big(F(z^*) - F(p_1^*)\big), \qquad \Pi_2^* = \big(1 - F(z^*)\big)(p_2^* - c),$$

where $p_2^* = (1 - \hat{q}_1^*)z^* + \hat{q}_1^* p_1^*$. The constraint (10) implies $\hat{q}_1^* = (c - J(z^*))/(p_1^* - J(z^*))$. Substituting the expression for $\hat{q}_1^*$ into the two profit functions gives

$$\Pi_1^* = \big(F(z^*) - F(p_1^*)\big)(p_1^* - c)\frac{c - J(z^*)}{p_1^* - J(z^*)}, \qquad \Pi_2^* = \big(1 - F(z^*)\big)(p_1^* - c)\frac{z^* - J(z^*)}{p_1^* - J(z^*)}.$$

Using $z^* - J(z^*) = \big(1 - F(z^*)\big)/f(z^*)$, we have

$$\Pi_1^* < \Pi_2^* \iff \big(1 - F(z^*)\big)^2 > \big(F(z^*) - F(p_1^*)\big)\big(c - J(z^*)\big)f(z^*).$$

It remains to establish this inequality.

Since z^* is Firm 2's unique best response, we have

$$\begin{aligned}\pi_2(p_1^*|p_1^*, \hat{q}_1^*) < \pi_2(z^*|p_1^*, \hat{q}_1^*) = \Pi_2^* &\iff \big(1 - F(p_1^*)\big)(p_1^* - c) < \big(1 - F(z^*)\big)(p_1^* - c)\frac{z^* - J(z^*)}{p_1^* - J(z^*)} \\ &\iff \big(1 - F(p_1^*)\big)f(z^*)\big(p_1^* - J(z^*)\big) < \big(1 - F(z^*)\big)^2,\end{aligned}$$

where we again use $z^* - J(z^*) = \big(1 - F(z^*)\big)/f(z^*)$. Therefore, it is enough to show that

$$\big(1 - F(p_1^*)\big)\big(p_1^* - J(z^*)\big) > \big(F(z^*) - F(p_1^*)\big)\big(c - J(z^*)\big). \tag{20}$$

After eliminating $\hat{q}_1$ from (10), Firm 1's reduced objective can be written as

$$\big(F(z) - F(p_1)\big)(p_1 - c)\frac{c - J(z)}{p_1 - J(z)}.$$

The first order condition with respect to p_1 at (p_1^*, z^*) gives

$$\big(F(z^*) - F(p_1^*)\big)\big(c - J(z^*)\big) = (p_1^* - c)f(p_1^*)\big(p_1^* - J(z^*)\big).$$

Hence (20) is equivalent to

$$\big(1 - F(p_1^*)\big)\big(p_1^* - J(z^*)\big) > (p_1^* - c)f(p_1^*)\big(p_1^* - J(z^*)\big).$$

Since $p_1^* - J(z^*) > 0$, this is equivalent to $1 - F(p_1^*) > (p_1^* - c)f(p_1^*)$, or equivalently $J(p_1^*) < c$. This inequality follows from regularity: because $p_1^* < z^*$ and $J(z^*) < c$, we have $J(p_1^*) < J(z^*) < c$. Thus $\Pi_2^* > \Pi_1^*$.

Part (iii). Observe that

$$\begin{aligned}
\Pi^m &= \hat{q}_1^*\big(1 - F(p^m)\big)(p^m - c) + (1 - \hat{q}_1^*)\big(1 - F(p^m)\big)(p^m - c) \\
&> \hat{q}_1^*\big(1 - F(p_1^*)\big)(p_1^* - c) + (1 - \hat{q}_1^*)\big(1 - F(z^*)\big)(z^* - c) \\
&= \big(F(z^*) - F(p_1^*)\big)\hat{q}_1^*(p_1^* - c) + \big(1 - F(z^*)\big)\big((1 - \hat{q}_1^*)z^* + p_1^*\hat{q}_1^* - c\big) \\
&= \Pi_1^* + \Pi_2^*,
\end{aligned}$$

where in the inequality we use the fact that p^m maximizes $\big(1 - F(p)\big)(p - c)$. Next,

$$BS = \int_{p_1^*}^{z^*} \hat{q}_1^*(\theta - p_1^*)dF(\theta) + \int_{z^*}^{1}(\theta - p_2^*)dF(\theta) > \int_{p^m}^{1}(\theta - p^m)dF(\theta) = BS^m,$$

since $p_2^* < z^* < p^m$ from the first part. Lastly,

$$\begin{aligned}
TS &= \int_{p_1^*}^{z^*} \hat{q}_1^*(\theta - c)dF(\theta) + \int_{z^*}^{1}(\theta - c)dF(\theta) \\
&> \int_{z^*}^{1}(\theta - c)dF(\theta) \\
&> -\big(1 - F(p^m)\big)c + \int_{p^m}^{1}\theta dF(\theta) = TS^m,
\end{aligned}$$

where the second inequality holds since $c < z^* < p^m$, which implies $\big(F(p^m) - F(z^*)\big)c < \int_{z^*}^{p^m} \theta dF(\theta)$. □

Appendix B Optimality of the no-subsidy condition

We allow Firm 1 to choose any $u = -t(0) \geq 0$ and show that $u = 0$ is optimal when Firm 1 offers a single option. Introducing u does not change the structure of Firm 1's optimal menu: it simply shifts the buyer's payoff from Firm 1 by the constant u, so the same arguments continue to apply. In particular, under positive equilibrium profits for both firms, it remains optimal for Firm 1 to offer a single option, which we summarize by a pair (a, b), where a is the lowest type served by Firm 1 and b is the quantity offered to all types above a. Moreover, $0 < a < z < 1$ holds in equilibrium; if instead $a \geq z$, then all types below z pay $-u$, so Firm 1 makes non-positive profit of $-F(z)u \leq 0$. Hence, given u, Firm 1 solves

$$\max_{0<a<z<1,\, 0<b<1} \int_0^z \left[-u + (\theta - c)x(\theta) - \int_0^\theta x(s)ds\right] dF(\theta) = \big(F(z) - F(a)\big)b(a - c) - F(z)u \tag{21}$$

subject to

$$\pi_2'(z|x, u) = -f(z)\big((1 - b)z + ab - c - u\big) + \big(1 - F(z)\big)(1 - b) = 0. \tag{22}$$

In (22), we use the relationship between z and Firm 2's posted price p_2:

$$\int_0^z x(s)ds + u = z - p_2 \iff p_2 = (1 - b)z + ab - u.$$

It must be the case that $b(a - c) > u$ and $a > c$, since otherwise the objective in (21) is non-positive. From (22), we have

$$b = \frac{c - J(z) + u}{a - J(z)}. \tag{23}$$

Now we plug (23) into (21), and Firm 1's problem becomes

$$\begin{aligned} \max_{u \geq 0} \max_{0<a<z<1} l(u; a, z) &:= \big(F(z) - F(a)\big)(a - c)\frac{c - J(z) + u}{a - J(z)} - F(z)u, \\ \text{s.t.} \quad 0 &< \frac{c - J(z) + u}{a - J(z)} < 1. \end{aligned} \tag{24}$$

Let (u^*, a^*, z^*) be a solution to (24). Suppose, toward a contradiction, that $u^* > 0$. By

the previous paragraph, $b^*(a^* - c) > u^*$ and $a^* > c$. The first order condition with respect to z can be written as

$$f(z)\big[b(a-c)-u\big] + \frac{\big(F(z)-F(a)\big)(a-c)J'(z)(c+u-a)}{(a-J(z))^2} = 0.$$

The first term is strictly positive. Hence the second term must be strictly negative, which implies $a > c + u$. Together with (23) and $b \in (0,1)$, this gives $J(z) < c + u < a$ and hence $a - J(z) > 0$.

The first order condition with respect to a is

$$\big(F(z)-F(a)\big)\big(c-J(z)\big) = (a-c)(a-J(z))f(a).$$

Since $a > c$ and $a - J(z) > 0$, the right-hand side is strictly positive. Therefore $J(z) < c$. Hence, holding (a^*, z^*) fixed, setting $u = 0$ remains feasible because

$$0 < \frac{c - J(z^*)}{a^* - J(z^*)} < 1.$$

It remains to show that $l(u; a, z)$ is strictly decreasing in u. Note that

$$\frac{\partial l(u;a,z)}{\partial u} = \frac{\big(F(z)-F(a)\big)(a-c)}{a-J(z)} - F(z).$$

This derivative is negative if and only if

$$F(z)\big(J(z)-c\big) < (a-c)F(a),$$

which is true since $J(z) < c$ and $a > c$. Thus lowering u^* to zero while holding (a^*, z^*) fixed strictly raises Firm 1's payoff, a contradiction. Therefore, it is optimal to set $u = 0$. □

Appendix C Equilibrium derivation in Section 4.3

For $\beta > 0$,

$$F_\beta(\theta) = 1 - (1-\theta)^\beta \qquad \text{and} \qquad J(\theta) = K\theta - (K-1),$$

where $K := \frac{\beta+1}{\beta} > 1$. Hence $J(\theta)$ is linear and strictly increasing. Let $\gamma := (1-z)/(1-p_1)$ and $d := \hat{q}_1 + (1-\hat{q}_1)K\gamma$. Using the constraint in (11) and dropping the constant $(1-c)^{\beta+1}$, the

problem can be written as

$$\max_{0<\hat{q}_1<1,\ 1/K<\gamma<1} \hat{q}_1(1-\gamma^\beta)(d-1)d^{-(\beta+1)}.$$

The first order conditions imply that, at any interior solution, the pair $(\hat{q}_1, \gamma)$ is determined by

$$\frac{1-\hat{q}_1}{\hat{q}_1} = \frac{\gamma^{(\beta-1)/2}}{\sqrt{K}}, \qquad \beta(1-\hat{q}_1)(K\gamma-1) = \hat{q}_1(1-\gamma^\beta). \tag{25}$$

Eliminating $\hat{q}_1$ yields

$$\beta(K\gamma-1)\gamma^{(\beta-1)/2} = \sqrt{K}\big(1-\gamma^\beta\big), \quad \gamma \in \Big(\frac{1}{K}, 1\Big). \tag{26}$$

Equation (26) has a unique solution $\gamma = \gamma_\beta \in (1/K, 1)$ since the left-hand side $L(\gamma) = \beta(K\gamma-1)\gamma^{(\beta-1)/2}$ is strictly increasing while the right-hand side $R(\gamma) = \sqrt{K}(1-\gamma^\beta)$ is strictly decreasing, with $L(1/K) = 0 < R(1/K)$ and $L(1) = 1 > 0 = R(1)$. Then $\hat{q}_1 = \hat{q}_{1,\beta} := 1/\Big(1+\frac{\gamma_\beta^{(\beta-1)/2}}{\sqrt{K}}\Big) \in (0,1)$ follows from (25).

Define

$$d_\beta := 1 + \frac{\hat{q}_{1,\beta}}{\beta}\big(1-\gamma_\beta^\beta\big).$$

Then the equilibrium variables are

$$p_1^* = 1 - \frac{1-c}{d_\beta}, \qquad z^* = 1-\gamma_\beta\frac{1-c}{d_\beta}, \qquad \hat{q}_1^* = \hat{q}_{1,\beta}, \qquad p_2^* = 1-\Big(\gamma_\beta + \hat{q}_{1,\beta}\big(1-\gamma_\beta\big)\Big)\frac{1-c}{d_\beta}.$$

Demands are

$$D_1 = \hat{q}_1^*\big[F(z^*) - F(p_1^*)\big] = \hat{q}_{1,\beta}\left(\frac{1-c}{d_\beta}\right)^\beta\big(1-\gamma_\beta^\beta\big),$$

$$D_2 = 1 - F(z^*) = \left(\gamma_\beta\frac{1-c}{d_\beta}\right)^\beta.$$

Profits are

$$\Pi_1^* = D_1 \times (p_1^* - c) = \frac{\hat{q}_{1,\beta}^2}{\beta}\big(1-\gamma_\beta^\beta\big)^2\left(\frac{1-c}{d_\beta}\right)^{\beta+1},$$

$$\Pi_2^* = D_2 \times (p_2^* - c) = \gamma_\beta^\beta\left[(1-\hat{q}_{1,\beta})(1-\gamma_\beta) + \frac{\hat{q}_{1,\beta}}{\beta}\big(1-\gamma_\beta^\beta\big)\right]\left(\frac{1-c}{d_\beta}\right)^{\beta+1}.$$

Lemma C.1. *Fix $c \in (0,1)$. In the equilibrium derived above, the following hold.*

(i) p_1^*, p_2^*, $\hat{q}_1^*$, D_2, and Π_2^* are strictly decreasing in β.

(ii) *As $\beta \to \infty$, both firms' demands and profits vanish:* $\lim_{\beta\to\infty} D_i = \lim_{\beta\to\infty} \Pi_i^* = 0,\ i = 1, 2.$

(iii) *As $\beta \downarrow 0$, Firm 2 serves the whole market:* $\lim_{\beta\downarrow 0} D_1 = \lim_{\beta\downarrow 0} \Pi_1^* = 0$, $\lim_{\beta\downarrow 0} D_2 = 1$, $\lim_{\beta\downarrow 0} \Pi_2^* = 1 - c$.

(iv) *D_1 and Π_1^* are strictly increasing for all sufficiently small $\beta > 0$ and strictly decreasing for all sufficiently large β. The ratio D_1/D_2 is strictly increasing in β.*

(v) *TS is strictly decreasing in β. BS is strictly increasing for all sufficiently small $\beta > 0$ and strictly decreasing for all sufficiently large β. Moreover,* $\lim_{\beta\downarrow 0} TS = 1 - c$, *while* $\lim_{\beta\downarrow 0} BS = \lim_{\beta\to\infty} BS = \lim_{\beta\to\infty} TS = 0$.

Proof. Write

$$s_\beta := \frac{1 - \hat{q}_{1,\beta}}{\hat{q}_{1,\beta}}.$$

Then the two first order conditions in (25) become $\gamma_\beta^\beta = K s_\beta^2 \gamma_\beta$ and $\beta s_\beta (K\gamma_\beta - 1) = 1 - \gamma_\beta^\beta$. Solving these two equations, and then using the definition of d_β, yields

$$K\gamma_\beta = \frac{1 + \beta s_\beta}{s_\beta(\beta + s_\beta)}, \qquad \gamma_\beta^\beta = \frac{s_\beta(1 + \beta s_\beta)}{\beta + s_\beta}, \qquad d_\beta = \frac{\beta + 1}{\beta + s_\beta}.$$

In particular, $0 < s_\beta < 1$. The positivity follows from $\hat{q}_{1,\beta} \in (0, 1)$, and $s_\beta < 1$ follows from $\gamma_\beta^\beta < 1$, since $\frac{s_\beta(1+\beta s_\beta)}{\beta + s_\beta} < 1 \Leftrightarrow s_\beta < 1$.

Since $K = (\beta + 1)/\beta$,

$$\gamma_\beta = \frac{\beta(1 + \beta s_\beta)}{s_\beta(\beta + s_\beta)(\beta + 1)}. \tag{27}$$

Equating its β-th power with the expression for γ_β^β above and taking logs gives

$$\beta \log\left(\frac{\beta(1 + \beta s_\beta)}{s_\beta(\beta + s_\beta)(\beta + 1)}\right) = \log\left(\frac{s_\beta(1 + \beta s_\beta)}{\beta + s_\beta}\right). \tag{28}$$

Let $G(\beta, s)$ denote the left-hand side of (28) minus the right-hand side. Differentiating G with respect to each argument and then evaluating at $s = s_\beta$ gives

$$G_s(\beta, s_\beta) = -\frac{\beta(\beta + 1)(1 + s_\beta)^2}{s_\beta(\beta + s_\beta)(1 + \beta s_\beta)} < 0,$$
$$G_\beta(\beta, s_\beta) = \log \gamma_\beta + \frac{1}{\beta + 1} + \frac{(\beta - 1)(s_\beta^2 - 1)}{(1 + \beta s_\beta)(\beta + s_\beta)}.$$

Using (27) and the expression for γ_β^β implies that

$$\frac{1}{\beta+1}+\frac{(\beta-1)(s_\beta^2-1)}{(1+\beta s_\beta)(\beta+s_\beta)}=1-\gamma_\beta+\frac{1-\gamma_\beta^\beta}{\beta(\beta+1)\gamma_\beta^\beta}.$$

Hence

$$G_\beta(\beta,s_\beta)=\log\gamma_\beta+1-\gamma_\beta+\frac{1-\gamma_\beta^\beta}{\beta(\beta+1)\gamma_\beta^\beta}.$$

Since $\gamma_\beta\in(0,1)$, set $\rho=-\log\gamma_\beta>0$. Then the last expression becomes

$$G_\beta(\beta,s_\beta)=\frac{e^{\beta\rho}-1}{\beta(\beta+1)}-\rho+1-e^{-\rho}.$$

The right-hand side is zero at $\rho=0$. Its derivative with respect to ρ is $\frac{e^{\beta\rho}}{\beta+1}-1+e^{-\rho}$. This derivative is strictly positive: by convexity of the exponential function, $\frac{e^{\beta\rho}}{\beta+1}+\frac{\beta e^{-\rho}}{\beta+1}>\exp\left(\frac{\beta\rho-\beta\rho}{\beta+1}\right)=1$, and therefore $\frac{e^{\beta\rho}}{\beta+1}-1+e^{-\rho}=\left(\frac{e^{\beta\rho}}{\beta+1}+\frac{\beta e^{-\rho}}{\beta+1}-1\right)+\frac{e^{-\rho}}{\beta+1}>0$. Thus $G_\beta(\beta,s_\beta)>0$. Since $G(\beta,s_\beta)=0$, the implicit function theorem gives

$$s'_\beta=-\frac{G_\beta(\beta,s_\beta)}{G_s(\beta,s_\beta)}>0.$$

Since $\hat{q}_1^*=1/(1+s_\beta)$, it follows immediately that $d\hat{q}_1^*/d\beta=-s'_\beta/(1+s_\beta)^2<0$. Hence the equilibrium cap is strictly decreasing in β.

Using the expressions above, the equilibrium prices can be written as

$$p_1^*=c+(1-c)\frac{1-s_\beta}{\beta+1},\qquad p_2^*=c+(1-c)\frac{1+\beta s_\beta}{(\beta+1)^2(1+s_\beta)}.$$

Since $s'_\beta>0$ and $0<s_\beta<1$,

$$\frac{d}{d\beta}\left(\frac{1-s_\beta}{\beta+1}\right)=-\frac{(\beta+1)s'_\beta+1-s_\beta}{(\beta+1)^2}<0.$$

Hence p_1^* is strictly decreasing in β.

We next show that p_2^* is also strictly decreasing. Let

$$B_\beta:=\frac{1+\beta s_\beta}{(\beta+1)^2(1+s_\beta)}.$$

Using $s'_\beta = -G_\beta(\beta, s_\beta)/G_s(\beta, s_\beta)$, we obtain

$$\frac{d}{d\beta}\log B_\beta = -\frac{2+(\beta-1)s_\beta}{(\beta+1)(1+\beta s_\beta)} + \frac{(\beta-1)s_\beta(\beta+s_\beta)}{\beta(\beta+1)(1+s_\beta)^3}G_\beta(\beta, s_\beta).$$

The first term is strictly negative because $0 < s_\beta < 1$. If $\beta \le 1$, the second term is non-positive, so the derivative is strictly negative. If $\beta > 1$, using $\log\gamma_\beta + 1 - \gamma_\beta < 0$,

$$G_\beta(\beta, s_\beta) = \log\gamma_\beta + 1 - \gamma_\beta + \frac{1-\gamma_\beta^\beta}{\beta(\beta+1)\gamma_\beta^\beta} < \frac{1-\gamma_\beta^\beta}{\beta(\beta+1)\gamma_\beta^\beta} = \frac{1-s_\beta^2}{(\beta+1)s_\beta(1+\beta s_\beta)}.$$

Therefore, for $\beta > 1$,

$$\frac{d}{d\beta}\log B_\beta < -\frac{2+(\beta-1)s_\beta}{(\beta+1)(1+\beta s_\beta)} + \frac{(\beta-1)(\beta+s_\beta)(1-s_\beta)}{\beta(\beta+1)^2(1+s_\beta)^2(1+\beta s_\beta)}.$$

It remains to show that the right-hand side is negative. This is equivalent to

$$(\beta-1)(\beta+s_\beta)(1-s_\beta) < \beta(\beta+1)(1+s_\beta)^2[2+(\beta-1)s_\beta].$$

Since $\beta > 1$ and $0 < s_\beta < 1$, the left-hand side is strictly smaller than $\beta(\beta+1)(1+s_\beta)$, while the right-hand side is strictly larger than $\beta(\beta+1)(1+s_\beta)$. Hence the inequality holds, and therefore $d\log B_\beta/d\beta < 0$. Thus B_β, and hence p_2^*, is strictly decreasing in β.

Now consider Firm 2's demand. Write

$$D_2 = \left(\gamma_\beta\frac{1-c}{d_\beta}\right)^\beta = [(1-c)R_\beta]^\beta, \qquad R_\beta := \frac{\beta(1+\beta s_\beta)}{s_\beta(\beta+1)^2}.$$

Differentiating gives

$$\begin{aligned}\frac{d}{d\beta}\log D_2 &= \log(1-c) + \log R_\beta + \beta\frac{d\log R_\beta}{d\beta} \\ &= \log(1-c) + \log R_\beta + \frac{2\beta s_\beta - \beta + 1}{(\beta+1)(1+\beta s_\beta)} - \frac{\beta s'_\beta}{s_\beta(1+\beta s_\beta)}.\end{aligned}$$

Since $c \in (0,1)$ and $s'_\beta > 0$, it is enough to show

$$\log R_\beta + \frac{2\beta s_\beta - \beta + 1}{(\beta+1)(1+\beta s_\beta)} < 0.$$

Because $R_\beta = \gamma_\beta/d_\beta$, $0 < \gamma_\beta < 1$, and $d_\beta = (\beta+1)/(\beta+s_\beta) > 1$, we have $0 < R_\beta < 1$. Using

$\log R < -\frac{2(1-R)}{1+R}$ for $R \in (0,1)$, it is enough to verify

$$\frac{2(1-R_\beta)}{1+R_\beta} - \frac{2\beta s_\beta - \beta + 1}{(\beta+1)(1+\beta s_\beta)} > 0.$$

Substituting $R_\beta = \beta(1+\beta s_\beta)/[s_\beta(\beta+1)^2]$, the left-hand side equals

$$\frac{2\beta^2 s_\beta^2 - \beta^2 + 5\beta s_\beta - 3\beta + s_\beta}{(\beta+1)(1+\beta s_\beta)(2\beta^2 s_\beta + 2\beta s_\beta + \beta + s_\beta)}.$$

The denominator is positive. Let $N_\beta(s) := 2\beta^2 s^2 - \beta^2 + 5\beta s - 3\beta + s$. Since $N_\beta'(s) = 4\beta^2 s + 5\beta + 1 > 0$, N_β is increasing in s. If $\beta \geq 1$, then $s_\beta \geq s_1 = 1/\sqrt{2}$, and hence $N_\beta(s_\beta) \geq N_\beta(s_1) = \beta\left(\frac{5}{\sqrt{2}} - 3\right) + \frac{1}{\sqrt{2}} > 0$. If $0 < \beta \leq 1$, then $s_\beta = \frac{\gamma_\beta^{(\beta-1)/2}}{\sqrt{K}} \geq \frac{1}{\sqrt{K}} = \sqrt{\frac{\beta}{\beta+1}}$. Let $\tilde{\beta} := \sqrt{\beta/(\beta+1)} \in (0,1)$. Then

$$N_\beta(s_\beta) \geq N_\beta(\tilde{\beta}) = 2\beta^2\tilde{\beta}^2 - \beta^2 + (5\beta+1)\tilde{\beta} - 3\beta = \frac{\tilde{\beta}\left[(1-\tilde{\beta})^3 + \tilde{\beta}^3(1+2(1-\tilde{\beta})^2)\right]}{(1-\tilde{\beta}^2)^2} > 0.$$

Thus $d\log D_2/d\beta < 0$, and hence D_2 is strictly decreasing in β. Since $\Pi_2^* = D_2(p_2^* - c)$, and both factors are positive and strictly decreasing in β, Π_2^* is strictly decreasing as well.

We now prove the limits for Firm 2's demand and profit. Since $D_2 = (\gamma_\beta(1-c)/d_\beta)^\beta$, with $\gamma_\beta \in (0,1)$ and $d_\beta > 1$, we have $0 < D_2 \leq (1-c)^\beta$. Hence $D_2 \to 0$ as $\beta \to \infty$. Since $p_2^* - c$ is bounded, $\Pi_2^* \to 0$ as well.

Now consider the case $\beta \downarrow 0$. Since $\gamma_\beta \in (1/K, 1) = (\beta/(\beta+1), 1)$, we have $\left(\frac{\beta}{\beta+1}\right)^\beta < \gamma_\beta^\beta < 1$. The lower bound converges to one as $\beta \downarrow 0$, and therefore $\gamma_\beta^\beta \to 1$. Also, $d_\beta = (\beta+1)/(\beta+s_\beta)$, so $1 < d_\beta^\beta < ((\beta+1)/\beta)^\beta \to 1$. Thus $D_2 \to 1$. Moreover, $d_\beta \to \infty$ as $\beta \downarrow 0$. Indeed, if $s_\beta \to a > 0$, then (27) gives $\gamma_\beta \to 0$, while (28) divided by β implies $\log \gamma_\beta \to a - 1/a$, a contradiction.[31] Hence $s_\beta \to 0$, so $d_\beta = (\beta+1)/(\beta+s_\beta) \to \infty$. The expression for p_2^* then gives $p_2^* \to 1$. Since $D_2 \to 1$, it follows that $\Pi_2^* = D_2(p_2^* - c) \to 1 - c$.

We next prove the limits for Firm 1's demand and profit. Recall that $D_1 = \hat{q}_{1,\beta}((1-c)/d_\beta)^\beta(1-\gamma_\beta^\beta)$. Since $0 < \hat{q}_{1,\beta} < 1$ and $d_\beta > 1$, we have $0 < D_1 \leq (1-c)^\beta \to 0$ as $\beta \to \infty$. As $\beta \downarrow 0$, we already showed that $\gamma_\beta^\beta \to 1$, and since $\hat{q}_{1,\beta}((1-c)/d_\beta)^\beta \leq 1$, it follows that $D_1 \to 0$. Finally, since $\Pi_1^* = D_1(p_1^* - c)$, with $0 < p_1^* - c < 1$, the two limits for D_1 imply

$$\lim_{\beta \downarrow 0} \Pi_1^* = \lim_{\beta \to \infty} \Pi_1^* = 0.$$

[31] Dividing (28) by β gives $\log \gamma_\beta = \beta^{-1} \log m_\beta$, where $m_\beta := s_\beta(1+\beta s_\beta)/(\beta + s_\beta)$. If $s_\beta \to a > 0$, then $m_\beta \to 1$ and $(m_\beta - 1)/\beta = (s_\beta^2 - 1)/(\beta + s_\beta) \to (a^2 - 1)/a = a - 1/a$. Using $\log r = r - 1 + o(r-1)$ as $r \to 1$, we obtain $\beta^{-1} \log m_\beta \to a - 1/a$.

We next prove the fourth statement. Note that

$$D_1 = \hat{q}_{1,\beta}\left(\frac{1-c}{d_\beta}\right)^\beta (1-\gamma_\beta^\beta) = \frac{\beta(1-s_\beta)(1-c)^\beta(\beta+s_\beta)^{\beta-1}}{(\beta+1)^\beta}. \tag{29}$$

Also, since $p_1^* - c = (1-c)(1-s_\beta)/(\beta+1)$,

$$\Pi_1^* = D_1 \frac{(1-c)(1-s_\beta)}{\beta+1}. \tag{30}$$

We first consider the case $\beta \downarrow 0$. We already showed above that $s_\beta \to 0$. Since $\gamma_\beta < 1$, (27) implies

$$\frac{\beta}{s_\beta} < \frac{(\beta+s_\beta)(\beta+1)}{1+\beta s_\beta} < (\beta+s_\beta)(\beta+1),$$

and therefore $\beta/s_\beta \to 0$. Also, $\gamma_\beta^\beta = (1+\beta s_\beta)/(1+\beta/s_\beta)$. Since $\beta/s_\beta \to 0$ and $\beta s_\beta = o(\beta/s_\beta)$, the expansion $\log(1+r) = r + o(r)$ gives $\log(\gamma_\beta^\beta) = -(\beta/s_\beta)(1+o(1))$. Hence $\log\gamma_\beta = -(1/s_\beta)(1+o(1))$. On the other hand, (27) and $\beta/s_\beta \to 0$ imply $\beta + s_\beta = s_\beta(1+o(1))$. Since also $1+\beta s_\beta = 1+o(1)$ and $\beta+1 = 1+o(1)$, we have $\gamma_\beta = \beta s_\beta^{-2}(1+o(1))$. Thus $\log\gamma_\beta = \log\beta - 2\log s_\beta + o(1)$. Equating the two expressions for $\log\gamma_\beta$ and multiplying by s_β gives $-1+o(1) = s_\beta\log\beta - 2s_\beta\log s_\beta + o(s_\beta)$. Since $s_\beta\log s_\beta \to 0$, it follows that $s_\beta\log(1/\beta)\to 1$.

We next evaluate $G_\beta(\beta, s_\beta)$. The results above imply $-\log\gamma_\beta \sim 1/s_\beta \sim \log(1/\beta)$.[32] Using the expression for $G_\beta(\beta, s_\beta)$, we have

$$G_\beta(\beta, s_\beta) = \frac{\exp\{\beta(-\log\gamma_\beta)\} - 1}{\beta(\beta+1)} + \log\gamma_\beta + 1 - \gamma_\beta.$$

Since $\beta(-\log\gamma_\beta)\to 0$ and $\beta(-\log\gamma_\beta)^2 \to 0$, the expansion $e^r - 1 = r + O(r^2)$ implies that the first term is $-\log\gamma_\beta + o(1)$. Moreover, $\log\gamma_\beta = -(1/s_\beta)(1+o(1))$ and $s_\beta \to 0$, so $\log\gamma_\beta \to -\infty$ and $\gamma_\beta = o(1)$. Hence $G_\beta(\beta, s_\beta) = 1 + o(1)$. Also,

$$G_s(\beta, s_\beta) = -\frac{\beta(\beta+1)(1+s_\beta)^2}{s_\beta(\beta+s_\beta)(1+\beta s_\beta)} = -\frac{\beta}{s_\beta^2}(1+o(1)),$$

where the last equality uses $\beta/s_\beta \to 0$. Therefore

$$s'_\beta = -\frac{G_\beta(\beta, s_\beta)}{G_s(\beta, s_\beta)} = \frac{s_\beta^2}{\beta}(1+o(1)) \quad \text{as } \beta \downarrow 0.$$

Next consider the case $\beta \to \infty$. Since $s'_\beta > 0$ and $0 < s_\beta < 1$, $s_\beta \to \bar{s}$ for some $\bar{s} \in (0, 1]$.

[32] For positive functions a_β and b_β, we write $a_\beta \sim b_\beta$ along a given limit if $a_\beta/b_\beta \to 1$ along that limit.

From (27), we can write $\gamma_\beta = (1 + 1/(\beta s_\beta))/[(1 + s_\beta/\beta)(1 + 1/\beta)]$. Hence $\beta \log \gamma_\beta = \beta \log(1 + 1/(\beta s_\beta)) - \beta \log(1 + s_\beta/\beta) - \beta \log(1 + 1/\beta) \to 1/\bar{s} - \bar{s} - 1$. Also, $\gamma_\beta^\beta = s_\beta(1 + \beta s_\beta)/(\beta + s_\beta) \to \bar{s}^2$. Hence $2 \log \bar{s} = 1/\bar{s} - \bar{s} - 1$. Since $s \mapsto 2 \log s - 1/s + s + 1$ is strictly increasing on $(0, 1]$, goes to $-\infty$ as $s \downarrow 0$, and equals 1 at $s = 1$, it follows that $\bar{s} \in (0, 1)$. Thus $-\log \gamma_\beta = O(\beta^{-1})$. Using the expressions for G_β and G_s, we obtain $G_\beta(\beta, s_\beta) = O(\beta^{-2})$ and $G_s(\beta, s_\beta) \to -(1 + \bar{s})^2/\bar{s}^2 < 0$.[33] Therefore

$$s'_\beta = O(\beta^{-2}) \quad \text{as } \beta \to \infty.$$

Now, since $D_1 > 0$, the sign of D_1' is the sign of its log-derivative. From (29),

$$\frac{d}{d\beta} \log D_1 = \frac{1}{\beta} - \frac{s'_\beta}{1 - s_\beta} + \log(1 - c) + \log \frac{\beta + s_\beta}{\beta + 1} + \frac{(\beta - 1)(1 + s'_\beta)}{\beta + s_\beta} - \frac{\beta}{\beta + 1}.$$

As $\beta \downarrow 0$, using $\beta/s_\beta \to 0$ and $s'_\beta = (s_\beta^2/\beta)(1 + o(1))$, multiplying this expression by β gives a limit equal to 1.[34] Hence $D_1' > 0$ for all sufficiently small $\beta > 0$. As $\beta \to \infty$, using $s_\beta \to \bar{s} \in (0, 1)$ and $s'_\beta = O(\beta^{-2})$, the log-derivative itself converges to $\log(1 - c) < 0$.[35] Hence $D_1' < 0$ for all sufficiently large β.

Similarly, since $\Pi_1^* > 0$, the sign of $(\Pi_1^*)'$ is the sign of its log-derivative. From (30),

$$\frac{d}{d\beta} \log \Pi_1^* = \frac{d}{d\beta} \log D_1 - \frac{s'_\beta}{1 - s_\beta} - \frac{1}{\beta + 1}.$$

As $\beta \downarrow 0$, the last two terms are negligible relative to $1/\beta$, so $\beta \, d \log \Pi_1^*/d\beta \to 1$. Hence $(\Pi_1^*)' > 0$ for all sufficiently small $\beta > 0$. As $\beta \to \infty$, the last two terms vanish, because $s'_\beta = O(\beta^{-2})$ and $1 - s_\beta \to 1 - \bar{s} > 0$. Therefore $d \log \Pi_1^*/d\beta \to \log(1 - c) < 0$, and hence $(\Pi_1^*)' < 0$ for all sufficiently large β.

Finally, we prove the monotonicity of D_1/D_2 and the welfare results. Since $\log \gamma_\beta + 1 - \gamma_\beta < 0$, the expressions for G_β and G_s above imply

$$0 < s'_\beta < \frac{(\beta + s_\beta)(1 - s_\beta)}{\beta(\beta + 1)^2(1 + s_\beta)} < \frac{s_\beta(1 - s_\beta)}{\beta(\beta + 1)},$$

where the last inequality follows from $\gamma_\beta < 1$ in (27), which is equivalent to $\beta + s_\beta < s_\beta(\beta +$

[33] Note that

$$G_\beta(\beta, s_\beta) = \log \gamma_\beta + 1 - \gamma_\beta + \frac{1 - \gamma_\beta^\beta}{\beta(\beta + 1)\gamma_\beta^\beta}.$$

Since $\gamma_\beta = 1 + O(\beta^{-1})$, we have $\log \gamma_\beta + 1 - \gamma_\beta = O(\beta^{-2})$. Also, $\gamma_\beta^\beta \to \bar{s}^2 \in (0, 1)$, so the last term is $O(\beta^{-2})$.

[34] Indeed, $\beta s'_\beta/(1 - s_\beta) = [s_\beta^2/(1 - s_\beta)](1 + o(1)) \to 0$, $\beta \log[(\beta + s_\beta)/(\beta + 1)] \to 0$, and $\beta(\beta - 1)(1 + s'_\beta)/(\beta + s_\beta) \to 0$.

[35] The rational terms vanish because $\log[(\beta + s_\beta)/(\beta + 1)] \to 0$, $(\beta - 1)(1 + s'_\beta)/(\beta + s_\beta) \to 1$, and $\beta/(\beta + 1) \to 1$.

$1)(1+s_\beta)$.

Using $D_1/D_2 = \beta(1-s_\beta)/[s_\beta(1+\beta s_\beta)]$ and the preceding bound, we obtain

$$\frac{d}{d\beta}\log\frac{D_1}{D_2} = \frac{1}{\beta(1+\beta s_\beta)} - \frac{1+\beta s_\beta(2-s_\beta)}{s_\beta(1-s_\beta)(1+\beta s_\beta)}s'_\beta > \frac{(1-s_\beta)^2}{(\beta+1)(1+\beta s_\beta)} > 0.$$

Since $D_1/\hat{q}_{1,\beta} + D_2 = 1 - F_\beta(p_1^*) = [(1-c)(\beta+s_\beta)/(\beta+1)]^\beta$,

$$\begin{aligned}\frac{d}{d\beta}\log\left(\frac{D_1}{\hat{q}_{1,\beta}} + D_2\right) &= \log(1-c) + \log\frac{\beta+s_\beta}{\beta+1} + \frac{\beta[1-s_\beta+(\beta+1)s'_\beta]}{(\beta+s_\beta)(\beta+1)} \\ &< \log(1-c) + \log\frac{\beta+s_\beta}{\beta+1} + \frac{1-s_\beta}{\beta+1} < 0,\end{aligned}$$

where the last inequality uses $\log r + 1 - r < 0$ for $r \in (0,1)$. Thus $D_1/\hat{q}_{1,\beta} + D_2$ is strictly decreasing in β.

To compare total surplus, let $u = 1 - F_\beta(\theta)$. Then $u \sim U[0,1]$, and the corresponding type is $\theta_\beta(u) = 1 - u^{1/\beta}$. A smaller u corresponds to a higher type. For each $u \in (0,1)$, $\theta_\beta(u)$ is strictly decreasing in β. The buyer receives one unit if $u \le D_2$, quantity $\hat{q}_{1,\beta}$ if $D_2 < u \le D_1/\hat{q}_{1,\beta} + D_2$, and zero otherwise. Since D_2, $D_1/\hat{q}_{1,\beta} + D_2$, and $\hat{q}_{1,\beta}$ are all strictly decreasing in β, the quantity received at each u cannot increase. Moreover, every trading type exceeds c, so a decrease in either the buyer's type or her quantity cannot increase total surplus. Hence TS is strictly decreasing in β.

For buyer surplus, integrating the buyer's equilibrium payoff, we have

$$BS = \frac{\left((1-c)/d_\beta\right)^{\beta+1}}{\beta+1}\left[\hat{q}_{1,\beta} + (1-\hat{q}_{1,\beta})\gamma_\beta^{\beta+1}\right].$$

Using $d_\beta = (\beta+1)/(\beta+s_\beta)$ and $\hat{q}_{1,\beta} = 1/(1+s_\beta)$, logarithmic differentiation gives

$$\frac{d}{d\beta}\log BS = \log\left[(1-c)\frac{\beta+s_\beta}{\beta+1}\right] + \frac{(\beta+1)(1+s'_\beta)}{\beta+s_\beta} - 1 - \frac{1}{\beta+1} - \frac{s'_\beta}{1+s_\beta} + \frac{d}{d\beta}\log\left(1+s_\beta\gamma_\beta^{\beta+1}\right).$$

Here $s_\beta\gamma_\beta^{\beta+1} = \beta s_\beta(1+\beta s_\beta)^2/[(\beta+1)(\beta+s_\beta)^2]$. Differentiating this expression and using the asymptotic results above shows that the last term is $O(1/s_\beta)$ as $\beta \downarrow 0$ and $O(\beta^{-2})$ as $\beta \to \infty$. Together with $s_\beta \log(1/\beta) \to 1$ and $s'_\beta \sim s_\beta^2/\beta$ as $\beta \downarrow 0$, and $s_\beta \to \bar{s} \in (0,1)$ and $s'_\beta = O(\beta^{-2})$ as $\beta \to \infty$, this yields

$$\lim_{\beta\downarrow 0}\frac{\beta}{s_\beta}\frac{d}{d\beta}\log BS = 1, \qquad \lim_{\beta\to\infty}\frac{d}{d\beta}\log BS = \log(1-c) < 0.$$

The first limit implies that $d\log BS/d\beta > 0$ for all sufficiently small $\beta > 0$, since $\beta/s_\beta > 0$.

The second implies that $d\log BS/d\beta < 0$ for all sufficiently large β. Since $BS > 0$, the log derivative has the same sign as BS'. Hence BS is strictly increasing for all sufficiently small $\beta > 0$ and strictly decreasing for all sufficiently large β.

The welfare limits follow from

$$0 \le BS \le TS \le \int_c^1 (\theta - c)\, dF_\beta(\theta) = \frac{(1-c)^{\beta+1}}{\beta+1}.$$

Since $TS = BS + \Pi_1^* + \Pi_2^*$, the profit limits proved above imply $TS \to 1 - c$ and $BS \to 0$ as $\beta \downarrow 0$. The upper bound also gives $BS \to 0$ and $TS \to 0$ as $\beta \to \infty$. □

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